AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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ABSTRACT

Title of research: **AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python**

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In response to escalating cyber threats and data insecurity, this thesis introduces the AC3 Algorithm, an encryption and decryption method. By blending the time-tested Caesar cipher and XOR operations into a five-level encryption process, the algorithm aims to significantly support data security. It emphasizes a user-friendly approach, targeting developers and tech companies to boost them with a practical tool for safeguarding sensitive data. Methodologically, the thesis follows a prototyping model, encompassing requirements gathering, rapid design, prototype development, user evaluation, refinement based on feedback, coding, rigorous testing, and ongoing maintenance. The thesis focuses on algorithm development, enhancing data security, developing a practical application, and advocating for strong digital security measures. While scalability testing and interface design receive less emphasis, the goal is to provide a comprehensive solution merging classic encryption methods into an accessible tool for securing information in an interconnected world. In essence, this thesis presents a tangible solution, a bridge between encryption techniques and practical implementation aimed at setting new standards in data security and privacy for broader accessibility and application.

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INTRODUCTION

Utilizing an algorithm combining five levels of encryption and decryption, applying both Ceasar cipher and XOR operations, in comparison to other program that supports encryption and decryption, our thesis establishes an easy-to-use application that contains a refined algorithm implementing Ceasar cipher and XOR operations through five levels, our algorithm aims to increase data security and data privacy to unusual levels. Our thesis, focusing on the development of a reliable encryption and decryption algorithm, addresses the pressing need for improved data security, privacy, and protection against cyber threats across various sectors of society. By enhancing the encryption process through a creative tandem of techniques, we aim to provide individuals and organizations with strong tool to safeguard their sensitive data, promoting a more secure digital environment in an increasingly interconnected society. This solution caters to individuals, organizations and even public authorities, collectively fostering a safer and more secure digital landscape, ultimately benefiting society at large.

Background of the problem

In the current digital era, the continuous threat of hacking poses a significant challenge, with numerous systems in the Philippines falling victim to data breaches. Notably, recent incidents, such as the compromise of a company's database, underscore the prevalent vulnerabilities in digital systems. The compromised data privacy of users is a major issue, particularly with the increasing interconnectivity of systems and the escalating volume of sensitive data shared across diverse sectors.

This scenario emphasizes the critical need for an innovative encryption and decryption solution to address the overarching problem of data security and privacy. Our focus is on providing a robust tool tailored for Developers who play a vital role in creating applications and projects that involve sensitive data. By implementing our AC3 Algorithm, Developers can fortify their systems against potential threats, contributing to an overall improvement in data security.

The AC3 Algorithm combines multiple encryption levels, including the Caesar Cipher and Xor operation offering an enhanced level of protection. This research aims to bridge the existing gaps in data protection by proposing a creative encryption method that safeguards sensitive data in applications or projects developed by our target audience. Through the adoption of our algorithm, we envision a substantial reduction in the risks associated with cyber threats and data breaches, ultimately contributing to the establishment of a more secure digital environment.

Overview of the current state of the technology

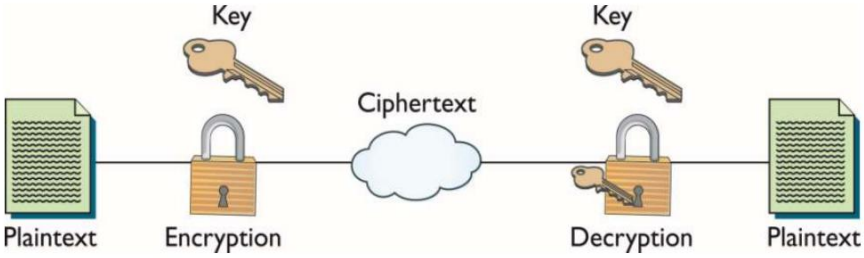


Figure 1: The process of Encryption and Decryption of plaintext

Classical Cryptography

Classical cryptography, rooted in the traditional cipher, involves scrambling plaintext to render it unreadable to unauthorized interceptors. The process necessitates a well-designed encryption and corresponding decryption process to ensure secure communication. With the advent of technology, particularly computers, the decryption of classical cryptography has become feasible. Advancements in computational technologies have not only facilitated decryption but have also provided diverse ways to generate, process, and utilize data (M. Iqbal Bhat & Kaiser J., 2021).

Modern Cryptography

The rise of computers has revolutionized cryptography, enabling the development of intricate and secure cryptographic techniques. Modern cryptography serves as a cornerstone of computer and communications security, producing practical end products. Despite leveraging computers in cryptographic algorithm creation, challenges persist. Researchers continually strive to innovate solutions, enhancing the security and privacy of digital communication and data (Mihir B., & Phillip R., 2005).

XOR Encryption Algorithm

The rise of computers has revolutionized cryptography, enabling the development of intricate and secure cryptographic techniques. Modern cryptography serves as a cornerstone of computer and communications security, producing practical end products. Despite leveraging computers in cryptographic algorithm creation, challenges persist. Researchers continually strive to innovate solutions, enhancing the security and privacy of digital communication and data (Mihir B., & Phillip R., 2005).

Advanced Encryption Algorithm (AES)

AES, or Advanced Encryption Standard, stands as a globally recognized and widely used cryptographic primitive. It serves as the gold standard for symmetric block ciphers, ensuring data security across applications (Bogdanov et al., 2019). However, even with its widespread adoption, AES is not without vulnerabilities. Weaknesses, such as the use of predictable or weak keys and initialization vectors, can significantly compromise AES security.

Post Quantum Cryptography

The impending development of quantum computers, with their unparalleled computational speed, poses a threat to existing public key cryptosystems. Post-Quantum Cryptography addresses this challenge by researching and advancing quantum-resistant primitives. The goal is to secure the current public key infrastructure against both quantum and classical computing threats, ensuring robust cryptographic systems in the face of evolving technological landscapes (National Institute of Standards and Technology, 2023).

Objectives of the study

The objective of this study centers on the development, implementation, and assessment of encryption and decryption solutions. This solution integrates the principles of the Caesar cipher and XOR operations. Through this strategic combination, the design endeavors to support data security and privacy. By employing the Caesar cipher and XOR algorithm in a designed application, the aim is to create a simple and easy to use app that demonstrates how the algorithm can be used in real life.

1. **Design** - implement and evaluate a encryption and decryption application.
2. **Incorporate** - Caesar cipher and XOR operation.
3. **Enhance** - data security and privacy through the creative encryption algorithm.
4. **Provide** - a potent tool for developers and tech companies to secure sensitive data.
5. **User Feedback** - from users through usability testing and surveys to evaluate the system's ease of use, reliability, and overall user satisfaction.

In ambition of achieving the stated objective, our study will focus on the design, implementation, and assessment of an innovative encryption and decryption algorithm.
This algorithm will intricately blend five levels of encryption and decryption techniques, combining the strength of Caesar cipher and XOR operations with supplementary enhancements. By doing so, we aim to markedly further strengthen data individuals and organizations to safeguard their sensitive information. To ensure its effectiveness, we will conduct thorough usability testing and surveys to evaluate the system's ease of user, reliability, and overall user satisfaction.

Scope and limitations of the study

SCOPE OF THE STUDY

- **Algorithm Development and implementation –** Focus on creating a proficient encryption and decryption algorithm using Caesar cipher, XOR operations, and additional enhancements.
- **Enhanced Data Security and Privacy –** Improve data security and privacy significantly by implementing the encryption process and quantify the level of enhancement.
- **Practical Application –** Develop a user-friendly encryption and decryption application to showcase the algorithm's real-world application.
- **Promotion of Digital Security –** Advocate for a more secure digital environment by providing a capable tool to safeguard sensitive data.

LIMITATIONS OF OUR ALGORITHM

- **Testing Boundaries and Capacity Constraints –** Focus on demonstrating the algorithm's functionality rather than extensively testing scalability in real-time or high-traffic scenarios.
- **User Experience and Interface Design –** Place less emphasis on user interface design, focusing primarily on the algorithm's functionality and effectiveness.
- **Specific Platform or Environment –** Develop the application for a specific platform or programming language, recognizing these limitations and aiming for a generalized understanding of the algorithm's potential across different platforms.
- **Insensitive to Input Distinction** - The algorithm lacks the capability to distinguish between plaintext and pre-encrypted text. Consequently, it may unintentionally process already encrypted input, potentially leading to redundancy or unexpected outcomes.
- **Algorithmic Constraints -** Due to the specific techniques employed, such as Caesar cipher and XOR operations, the algorithm may exhibit limitations in

handling certain types of data or fail to provide optimal security in scenarios with advanced cryptographic requirements.

- **Sensitivity to Key Management -** The security of the algorithm heavily relies on effective key management practices. Weaknesses in key generation, distribution, or storage could compromise the overall robustness of the encryption.
- **Limited Resistance to Cryptanalysis -** While designed to enhance data security, the algorithm may exhibit susceptibility to certain cryptographic attacks. The level of resistance to cryptanalysis may be influenced by factors such as key length and algorithmic vulnerabilities.

LITERATURE REVIEW

Review of related literature, studies or systems

Local

Hybrid Encryption Algorithm Implementation on Electronic Mail Service

A study by Calinawan, C. (2015) aims to develop a hybrid encryption algorithm by combining RSA and Solitaire Encryption Techniques, enhancing reliability, speed, and security against potential attacks. It introduces a Blu€ key generation technique to conceal the session key, reducing vulnerability to hackers. The goal is to integrate this new algorithm into an open-source email client, potentially revolutionizing email security for exchanging encrypted text messages over the internet. This effort aligns with creating a more robust Pretty Good Privacy mechanism, enabling secure communication. The amalgamation of RSA and Solitaire Encryption forms the foundation for this innovation, while the Blu€ key generation technique enhances security. Overall, the study intends to offer a fresh approach to encrypted email communication, emphasizing heightened security measures. The amalgamation of well-known encryption methods and the introduction of a new key generation technique are pivotal aspects of this research. Ultimately, the objective is to improve encryption methods and fortify online communication against potential threats.

SMS Encryption-Decryption for Nokia Series60 2nd Edition Mobile Phones

A study by Joy, R., Lim, O., & Ryan, D. (2013) aims to develop a SMS, known as Short Message Service, offers cost-effective communication with instant delivery. It facilitates text message exchange via mobile devices, supporting unified messaging for various message types like voicemail, email, and fax. This study primarily focused on bolstering security for messages transmitted through mobile devices, particularly emphasizing SMS encryption for enhanced mobile communication security. Due to SMS vulnerabilities over the GSM network, the essay recommended additional encryption measures. It highlighted

comparisons among different encryption algorithms for securing SMS transfers. The study outlined the design and implementation of a proposed application tailored for N-series mobile phones. This app encrypts and decrypts SMS using three distinct algorithms, aiming to maximize message security for users. It concluded with prototypes showcasing secured SMS and proposed future extensions for the application.

Implementing Symmetric Cryptography on SMS Messaging for Generic Phone Using J2ME Platform

A study by Barrios, E. T., Avellanosa, M. M. F., Canuday, J. M. V., & Uy Jr., R. L. (2011) aimed to implement cryptography on common mobile phones, providing users with a tool to enhance the security and privacy of transmitted or received messages. Focusing on applying the Data Encryption Standard (DES) algorithm for encrypted text exchange via SMS, this research project centered on implementing symmetric cryptography within the J2ME Platform for generic phones. The primary goal was to design a mobile phone application enabling encrypted text exchange, requiring both phones to have encryption software installed for message encryption and decryption using a password. The researcher utilized experimental development with a descriptive technique to achieve the established goals, emphasizing the practical implementation of symmetric cryptography for SMS encryption on widely used mobile devices.

Implementation of Advanced Encryption Standard Cryptographic Algorithm in a Test Bank System

A study by Lampel B. Berol, & Ritchie A. Reyna. (2018) aimed to fortify information security against hackers by employing cryptography, enhancing system sensitivity to unforeseen attacks, reliability, and processing speed. The objective involved crafting a test banking system integrating cutting-edge encryption algorithms for secure data creation and retrieval. Primarily, the focus was on developing a comprehensive library of test questions structured around Bloom's Taxonomy's six levels. To achieve this, the study followed agile development principles, emphasizing adaptability to rapidly changing conditions,

accommodating user demands, meeting accelerated project deadlines, and addressing similar dynamic challenges. The core aim was to leverage cryptography in fortifying data security within the test banking system while employing an agile approach to ensure flexibility and responsiveness to evolving needs and constraints.

ANDROID MULTIMEDIA CIPHER USING HIERARCHICAL CLUSTERING ALGORITHM

A study by Ferrera, J. C., Carlos, J., Calamlam, D. R., & Kristel, D. (2016) aimed to enhance multimedia file security for Android users by employing the hierarchical clustering technique to alter file content. The program's foundation relied on hierarchical clustering as its core framework, incorporating principles from the Disk Encryption Theory and Chaos Theory. Using these theories, the software encrypted multimedia files by modifying their content, ensuring security. Subsequently, it could decrypt the files by accurately restoring their original content. This process involved leveraging hierarchical clustering to manipulate multimedia content, providing an added layer of security through content alteration and restoration, based on principles from Disk Encryption Theory and Chaos Theory.

Universal Intelligent Data Encryption Standards: A Review

A study by Ravi, R. V., & R. Mahalakshmi. (2014) Aimed to develop in the realm of electronic communication, safeguarding data stands out as one of the most formidable challenges. The significance of secure data exchange across the internet and various media continues to rise steadily. Within security communities, a pivotal pursuit revolves around finding optimal solutions to offer essential protection against intruders' attacks while ensuring timely service provision. Cryptography emerges as a key player in data security, transforming information from its original state into an unreadable format. Two primary criteria distinguish encryption systems from one another: their capacity to shield protected data against cryptanalytic attacks and their speed and efficiency in executing this process. Hence, cryptography assumes a critical role in fortifying data security, tackling the ever-

evolving challenges posed by intruders and ensuring efficient data protection in electronic communication mediums.

Implementation of Euclidean Algorithm for Single Key Authentication System

A study by Margate, R. K., Maturan, C., Rosales, R. P., & Gupit, D. (2019). Aimed to develop a rapidly advancing world of communication and technology, securing information is paramount for crucial national interests. Encryption serves as a powerful tool in fortifying password-based authentication, especially when integrated with other robust information security measures. A pivotal recommendation involves revising the current login system policy to actively advocate and incentivize the widespread adoption of security keys and encrypted passwords. This shift aims to prioritize user account and information protection while ensuring alignment with legitimate coverage functions essential for robust information security. The vulnerability of most password authentication systems lies in the reliance on a single secret phrase. If this passphrase is compromised, adversaries gain the ability to impersonate victims, accessing their entitled resources. Therefore, promoting the adoption of security keys and encrypted passwords becomes imperative to mitigate the risks associated with potential breaches in password-based authentication systems.

Security and File Privacy Management Application for Flash Drives

A study by Manlangit, M. G., & Ve, J. (2013). Aims to address the vulnerability of flash drive contents by establishing security measures that restrict unauthorized access and limit the spread of sensitive information. With flash SSDs being a prevalent technology across various sectors, their lack of inherent security poses risks. Many users have experienced compromised accounts, leading to unauthorized access to their flash drive files. To combat this issue, advocates propose the development of a "Security and File Privacy Management Application for Flash Drives." This application seeks to mitigate the proliferation of hazardous files and prevent unauthorized users from accessing sensitive data stored on flash drives. The goal is to enhance security measures for these widely used storage devices

and prevent unauthorized access, thus safeguarding user data and reducing potential risks associated with compromised accounts.

Development of a Secured Web Page of CAAP SIS using Cryptography

A study by Toria, A. (2014) aims of the expansion of information technology has exponentially increased user risks, necessitating robust security measures. The primary objective is to render communication nearly impossible to decrypt without the corresponding key, utilizing state-of-the-art encryption software. This study concentrates on addressing security challenges by offering solutions concerning integrity, non-repudiation, authorization, and confidentiality. Dealing with hacker attacks involves preventing unauthorized access, while facing legal attacks requires concealing data existence, as legal systems possess means to subpoena data and demand disabling protective measures for data retrieval. The study's findings underscored the promising potential for enhanced security through the development of an audit trail. This mechanism holds significant promise in bolstering security measures, providing an additional layer of protection against unauthorized access and ensuring the integrity and traceability of data interactions.

Systems Development for Records Archiving and Digital Documents Repository: A Case Study

A study by Gamba, M. P., Ocbian, M. M., & Gamba, M. N. (2014) aims in response to the pressing need for efficient data access and exchange, the research aimed to transcend the limitations of traditional server-client document transfers by establishing records archiving and repositories for the geographically dispersed Sorsogon State College campuses in the Philippines. Addressing communication challenges between these campuses, a General User Interface (GUI) application was developed, offering accessibility via web-enabled computers and mobile devices without requiring specific installations, utilizing web browsers for compatibility across diverse platforms. By deploying web-based archiving and repositories on the main campus and connecting satellite campuses through a virtual

private network (VPN) atop existing Internet service providers, the system facilitated constructive connectivity. This integration fosters seamless communication and real-time information flow, overcoming geographic barriers to create a more interconnected and accessible data environment across the college's campuses.

A Novel ASCII Code-based Polybius Square Alphabet Sequencer as Enhanced Cryptographic Cipher for Cyber Security Protection (APSAlpS-3CS)

A study by Arroyo, J. C. T., Roy, A., & Jhone, A. (2020) aims of the frequency analysis showcases the enhanced security of the improved Polybius cipher, offering a robust transformation of plaintext into ciphertext that poses a significant challenge to decryption. In this cipher, each encrypted letter undergoes alteration by shifting cell components via a secret key and their corresponding ASCII decimal values. This technique results in elements that span a broader range of potential values, contributing to heightened encryption strength. However, owing to the increased complexity in both encryption and decryption procedures, the execution time extends to 0.0031 milliseconds, a trade-off for the heightened security measures implemented in this cipher.

A unique message encryption technique based on enhanced blowfish algorithm.

A study by Dulla, G. L., Gerardo, B. D., & Medina, R. P. (2019) aims in the realm of online transactions and message transmission, simplicity has inadvertently led to increased vulnerability and susceptibility to manipulation. Consequently, ensuring security becomes paramount to safeguarding site content and thwarting unwanted intrusion attempts. This study focuses on employing an encryption method rooted in an enhanced version of the Blowfish algorithm to fortify both plaintext and file-based message data. The improvements made to the Blowfish algorithm involve optimizing its performance by reducing the number of rounds, extending block lengths, and incorporating additional transformation techniques within specific rounds during encryption and decryption processes. The investigation utilized an application operating on an x64 system via Visual

Studio, aiming to enhance encryption strength and security measures to counteract the evolving challenges posed by online data manipulation and unauthorized access attempts.

Computational Complexity of Modified Blowfish Cryptographic Algorithm on Video Data

A study by Adeniyi, A. E., Misra, S., Daniel, E., & Bokolo, A. (2022) aims in Cryptography serves as a fundamental tool in safeguarding consumer information, with the Blowfish encryption method standing out for its simplicity, robustness, and effectiveness. Particularly for data and applications demanding high-level security, utilizing Blowfish is recommended. However, concerning the findings from the Avalanche Effect analysis, the standard Blowfish algorithm showcased superior security across all categories of video file sizes compared to the modified Blowfish method. Despite the modifications aimed at enhancing security, the standard algorithm demonstrated more robust encryption capabilities, especially when applied to various video file sizes.

A CCA-PKE Secure-Cryptosystem Resilient to Randomness Reset and Secret-Key Leakage

A study by Labao, A., & Adorna, H. (2022) aims in the realm of public-key cryptosystems, recent advancements have incorporated novel security considerations. Notably, these advancements confront adversaries wielding a combination of attacks—accessing secret-key leakage queries alongside additional encryption queries using their chosen randomness. This scenario implies interference during both encryption and decryption processes under this security paradigm. The research endeavors to tackle the challenge posed by combined randomness and secret-key leakage attacks. The focus lies in developing solutions to mitigate the potential risks and vulnerabilities arising from these simultaneous attack vectors, aiming to fortify the security of public-key cryptosystems against such sophisticated threats.

Foreign

Importance of Cryptography in Information Security

A study by Gençoğlu, M. (2019) emphasizes the dominance of online communication in the contemporary world. Despite this prevalence, many users remain unaware of the inherent risks involved. The exchange of private information over the internet exposes data to various potential attacks, with cybercriminals adeptly targeting sensitive details such as passwords, credit card numbers, and personal addresses. Encryption stands as a crucial defense mechanism, converting information into an unintelligible format for protection. However, as attackers evolve in sophistication, Gençoğlu (2019) argues that relying solely on cryptography may not suffice to fully safeguard information. The study asserts the necessity for modern cryptography to furnish robust protection and digital keys, ensuring the integrity and security of information transmitted in cyberspace. Enhancing encryption and decryption techniques becomes imperative to afford genuine internet users the highest attainable level of security. In this digital era, the challenge lies in outmaneuvering cybercriminals to grant both senders and recipients peace of mind. It is crucial to transform information into unreadable forms to ensure its protection and safe delivery to its intended destination.

CRYPTOGRAPHY: THE EMERGING TECHNOLOGY

A study by Nadiminti, S., Kola, C. P., & Shankar, P. B. (2021) highlights the evolution of cybercriminals' tactics, demonstrating increased proficiency in stealing online information. The susceptibility of traditional cryptographic methods to decryption, owing to modern cryptanalysis techniques, raises concerns about potential attacks on shared information when relying on outdated encryption approaches. Thus, the use of obsolete encryption methods for information security is strongly discouraged. Poornima, N et al (2021) contribute to the discourse by emphasizing the subjective nature of "security" perceptions and acceptable risk levels. Ongoing advancements and innovations in security methodologies are noted for fortifying data protection. The continual evolution and persistent application of cryptography in secure environments emerge as a favored choice.

The primary goal of cryptography remains thwarting unauthorized access and data breaches, serving as an enduring tool against hacking and unauthorized data access.

COMBINATION OF CAESAR CIPHER ALGORITHM AND RIVEST SHAMIR ADLEMAN ALGORITHM FOR SECURING DOCUMENT FILES AND TEXT MESSAGES

A study by Gunawan, I., et al (2019) merges the Caesar Cipher Algorithm with the RSA algorithm. Caesar Cipher, an ancient algorithm, employs a substitution cipher method by shifting all plaintext characters using a consistent shift value. However, Caesar Cipher's weakness lies in its vulnerability to brute force attacks, enabling the retrieval of the original message. The combination of Caesar Cipher with the RSA algorithm aims to address this vulnerability. By integrating Caesar Cipher with RSA, the weaknesses of the former are mitigated. This amalgamation offers an enhanced data security system compared to the use of either method in isolation. The utilization of calculations based on alphabetic structures, alongside the incorporation of prime numbers inherent in the RSA algorithm, serves to strengthen the overall data security system.

Developing a Modified Hybrid Caesar Cipher and Vigenere Cipher for Secure Data Communication

A study by Omolara et al (2014) aims to contribute to classical cryptography by introducing a novel modified hybrid encryption method for plaintext. While security can be bolstered by employing big key spaces, numerous rounds, and complex processes, these factors might also slow down operations. In this work, a modified hybrid of the Caesar and Vigenere ciphers is proposed, offering a blend of dissemination and confusion that surpasses typical Classical ciphers. The modifications made to the Caesar and Vigenere ciphers involve enlarging and altering them to incorporate alphabets, numerals, and symbols. Additionally, complete conflation and diffusion mechanisms were incorporated. This study amalgamates contemporary cipher attributes into classical ciphers, creating a

highly robust encryption scheme that presents challenges for decryption techniques like frequency analysis and brute force methods. The resultant modified Caesar Cipher and Vigenere Encryption hybrid represents a fusion of classical and contemporary cipher qualities, strengthening the overall security and rendering it significantly challenging to crack.

IMPLEMENTATION OF COMBINATION VIGENERE CIPHER AND RSA IN HYBRID CRYPTOSYSTEM FOR TEXT SECURITY

A study by Jamaludin, J. (2020) aims to address inherent flaws in two forms of cryptography by devising and implementing a solution through a hybrid cryptosystem coding technique. By merging the Vigenere Cipher for message encoding and the RSA cryptographic technique for key encoding, this hybrid approach aims to rectify vulnerabilities present in both cryptographs, consequently bolstering encryption security. However, as observed in the study's findings, this amalgamation results in increased encryption and decryption durations. Specifically, encrypting 59,400 characters necessitates 424 milliseconds, while decrypting the same amount requires 335 milliseconds. Despite the heightened security afforded by this hybrid approach, the trade-off involves longer processing times for encryption and decryption procedures.

Modified Vegenere Cipher to Enhance Data Security Using Monoalphabetic Cipher

A study by Agustini et al. (2019) emphasizes the significance of swift data communication in the context of public networks for ensuring information security. Within this domain, various cipher techniques, including both mono- and poly-alphabetic ciphers, play a crucial role in enhancing security. Despite its widespread use, the Vigenere cipher is found to have vulnerabilities due to its reliance on additive cipher computations. To strengthen this cryptographic approach, researchers suggest merging the Vigenere cipher with the Affine cipher, known for its two-step encryption method involving both additive and multiplicative ciphers. This fusion, implemented through Matlab simulations, aims to increase complexity and thwart potential attacks. Evaluation tools such as Vigenere

Analyzer and Analysis highlight how this innovative approach surpasses the vulnerabilities of the Vigenere Cipher, presenting a more sophisticated Formulalgorithm. The combination of these classical ciphers results in a heightened level of security, establishing a novel encryption approach that outperforms conventional methods, especially in safeguarding against single-letter substitution attacks.

Formulation of an Improved Hybrid Cipher System

The study conducted by Ahmadrufa'i et al. (2020) addresses the critical need for data security in the contemporary era, where the extensive use of the internet has led to a surge in data transfer activities. Recognizing the vulnerability of information shared over the internet to various attacks, the researchers propose a modification and amalgamation of three ciphers: RSA model, Hill Cipher, and Shifting Cipher. The objective is to rectify the shortcomings of conventional ciphers, including but not limited to the Caesar Cipher, Hill Cipher, Vigenere Cipher, and Railfencing Cipher. The abstract of the study emphasizes the importance of data security in the current landscape, highlighting the frequent transfer of sensitive information over the internet. Cryptography is introduced as a solution to the vulnerabilities associated with information sharing, wherein the plaintext message is encoded to ciphertext. Despite the existence of various encryption techniques such as Caesar cipher, Hill cipher, Vigenere cipher, Railfencing cipher, and Playfair cipher, conventional methods have demonstrated weaknesses susceptible to brute force attacks and traditional cryptanalysis. In their work, Ahmadrufa'i et al. (2020) specifically focus on three existing ciphers: Caesar cipher, Hill cipher, and RSA Algorithm. Through a rigorous analysis, the weaknesses inherent in these ciphers are demonstrated, paving the way for the exploration of an innovative approach. The study employs the concept of functions (bijection) and leverages the idea of the composition of functions to combine and compose the functions of the three ciphers. This results in the development of an improved encryption technique, called a hybrid cipher system, achieved through RSA Algorithm with Shifting and Hill ciphers. The research contributes to the ongoing discourse on data security, offering a nuanced perspective on the weaknesses of existing ciphers and presenting a hybrid solution to enhance encryption techniques. The findings of this study

are anticipated to provide valuable insights into the continual evolution of cryptographic methods and their applications in safeguarding sensitive information in the digital age.

A new Modified method of Cryptography using Caesar Cipher

In their 2021 study, A. K. Tripathi et al. highlight the critical importance of data security in modern internet-based communication. Emphasizing the sender, medium, and receiver components, the study underscores the necessity of encryption for secure data transmission over the internet, especially in sectors like corporate, banking, and government where data sharing is prevalent. The authors focus on encryption techniques, with a particular spotlight on the Caesar cipher. They analyze the Basic Caesar cipher, Delta formation Caesar cipher, and XOR Caesar cipher, considering parameters such as Avalanche Effect, Frequency Test, and susceptibility to Brute Force attacks. A notable contribution of the study lies in the authors' attempt to modify the Caesar cipher method to produce readable ciphertext. This modification aims to enhance data security while minimizing suspicion during cryptanalysis. By presenting ciphertext that can be read, the study introduces a novel approach to fortify the security of the Caesar cipher against potential decryption attempts. In conclusion, A. K. Tripathi et al.'s study provides insights into data security challenges in internet-based communication. The focus on the modification of the Caesar cipher adds a layer of security, offering a pragmatic solution to safeguard transmitted data in the contemporary digital landscape.

Enhancing the Security of Caesar Cipher Substitution Method using a Randomized Approach for more Secure Communication

A study by Jain et al. (2015) assesses the limitations and vulnerabilities inherent in traditional encryption algorithms such as the Caesar cipher and the Transposition cipher. To address these flaws, a modified Caesar algorithm is proposed, aiming to rectify the restrictions of the original Caesar cipher. Enhancing the strength of this method involves a fusion of a double columnar transposition technique with a randomized substitution

strategy. Cryptanalysis reveals that decrypting this modified method using frequency analysis becomes impractical. A brute force attack on this technique would require attempting a staggering 256 distinct key combinations, rendering it exceedingly challenging to crack. Further fortification of this algorithm's security could be achieved by combining it with other encryption techniques or by transitioning from a symmetric key approach to an asymmetric key method. These strategic enhancements bolster the encryption's resilience against potential decryption attempts, emphasizing the robustness of this modified algorithm.

DESIGN, ANALYSIS, AND IMPLEMENTATION OF A NEW LIGHTWEIGHT BLOCK CIPHER FOR PROTECTING IOT SMART DEVICES

A study by Ragab et al. (2021) introduces a modified version of the XXTEA symmetric block cipher, M-XXTEA, designed to secure low-resource devices like IoT gadgets, providing private data transmission over the internet. Addressing vulnerabilities in the original XXTEA, the M-XXTEA iteration thwarts chosen-plaintext and key-related attacks, enhancing security. Integrating a chaotic key generating system, it constantly alters cipher keys for each data block encrypted, akin to a one-time pad, ensuring additional protection. M-XXTEA's versatility accommodates various text block and key sizes while significantly improving data security. Performance evaluations against original XXTEA and AES demonstrate M-XXTEA's superiority, surpassing AES by 60% in encryption and decryption time efficiency and exhibiting a 57% speed advantage. Positioned as an effective security solution, M-XXTEA ensures robust data protection, making it suitable for smart cities, e-health systems, and industrial IoT devices.

AN OVERVIEW AND ANALYSIS OF HYBRID ENCRYPTION: THE COMBINATION OF SYMMETRIC ENCRYPTION AND ASYMMETRIC ENCRYPTION

A study by Qixin, Z. (2021) states that the evolution of cryptographic algorithms has notably bolstered information security, with symmetric and asymmetric encryption

standing as the primary categories in cryptography. While symmetric encryption offers rapid data processing and is ideal for encrypting large volumes of data, its security is not as robust as asymmetric encryption, primarily due to its reliance on a single pair of keys. However, securing the key can significantly enhance its security. An approach gaining traction involves encrypting the message using a symmetric technique while safeguarding the key with an asymmetric algorithm. This study delves into information security concerns and emphasizes the burgeoning role of hybrid encryption algorithms, foreseeing their extensive utilization in the future. The amalgamation of encryption algorithms is poised to enhance transmission security without exacerbating additional challenges.

A HYBRID ENCRYPTION SOLUTION TO IMPROVE CLOUD COMPUTING SECURITY USING SYMMETRIC AND ASYMMETRIC CRYPTOGRAPHY ALGORITHMS

A study by Abroshan, H. (2021) proposes an efficient cryptographic approach to bolster cloud computing security while maintaining optimal performance. Given the crucial role of computational speed in cloud environments, complexity in cryptography might not be advantageous. Hence, this method merges an elliptic-curve-based technique with an enhanced Blowfish algorithm. By encrypting data using the Blowfish algorithm and employing the elliptic curve method for key encryption, this approach enhances security and performance. Additionally, to guarantee data integrity, a digital signature mechanism is employed. Evaluation demonstrates improved throughput, execution speed, and memory consumption characteristics, affirming the efficacy of this approach in fortifying cloud computing security without compromising performance.

A Modified Advanced Encryption Standard for Data Security

A study by Abikoye et al. (2019) delves into the exploration of Hadoop technology by establishing an experimental platform. Initially showcasing conventional data encryption techniques, the study then proposes a modified encryption algorithm. This algorithm integrates random disturbance information to fortify data security, targeting the

vulnerabilities inherent in traditional methods within the cloud computing environment. Experimental findings substantiate the efficacy of the suggested technique for encryption within the context of cloud computing, highlighting its suitability and potential in bolstering data security in this domain.

AES-RC4 ENCRYPTION TECHNIQUE TO IMPROVE FILE SECURITY

A study conducted by Nur, A. et al. (2019) underscores the critical importance of data security in public networks, given the looming threat of unauthorized data theft. Cryptography emerges as a pivotal means to safeguard data, involving two fundamental operations: encryption and decryption. Encryption converts plaintext into ciphertext, and decryption reverses this process. The level of security in encryption techniques is directly linked to their complexity; generally, more intricate algorithms are more resistant to unauthorized access. This study proposes the amalgamation of RC4 and AES algorithms with the objective of reinforcing the security of sensitive information. This combination aims to heighten the algorithm's complexity and robustness. The research assesses the algorithm's security through the avalanche effect (AE), wherein a modification in a key bit is used to measure the AE value. A larger discrepancy in the results signifies improved encryption outcomes, indicating heightened security measures.

A COMPLEX POLYALPHABETIC CIPHER TECHNIQUE MYANMAR POLYALPHABETIC CIPHER

A study by Aung, T. M. (2019) emphasizes the importance of information security in storing and transmitting data over public communication networks has surged alongside the rapid growth of digital information exchange. Cryptography has emerged as a crucial tool in safeguarding data from malicious attacks, utilizing techniques like monoalphabetic and polyalphabetic ciphers to ensure data confidentiality. These methods aim to conceal plaintext messages behind encrypted text, rendering them indecipherable to potential attackers. The Vigenère cipher, a popular polyalphabetic encryption method, traditionally employs a table to encrypt English alphabets. However, contemporary analyses suggest

that Vigenère encryption is no longer vulnerable to attacks like Kasiski and Friedman. To further fortify data security in public network communication and storage, researchers propose a new polyalphabetic cipher in this paper. This enhanced encryption technique combines the Vigenère and Affine ciphers, offering diffusion and confusion properties for added security. Additionally, to bolster data security for the Myanmar language, the proposed approach expands to create an encryption method based on Myanmar alphabet letters. Utilizing Myanmar alphabet characters as the key and transforming plaintext messages into encrypted Myanmar alphabet characters, this cipher technique aims to fortify data security in Myanmar language communications.

A NEW MODIFIED CAESAR CIPHER CRYPTOGRAPHY METHOD WITH LEGIBLE CIPHERTEXT FROM A MESSAGE TO BE ENCRYPTED

A study by Purnama and Rohayani (2015) addresses the concealment of information through cryptography and steganography. They highlight the vulnerability of encrypted messages in traditional cryptography, which may be deciphered by cryptanalysts. The suspicion aroused by unreadable text messages prompts decryption attempts. In contrast, steganography introduces challenges in determining the method or pattern of composing hidden messages. The authors propose a modification to the Caesar cipher method, aiming to produce ciphertext that can be read. This modification involves replacing the alphabet, dividing it into two parts: vowels replaced with corresponding vowels, and consonants replaced likewise, with select consonants remaining unchanged due to their infrequent usage in Indonesian text. The outcome is a readable ciphertext, strategically designed to reduce suspicion during cryptanalysis. The study demonstrates how modifying established cryptographic methods can enhance security and complicate decryption attempts. Purnama and Rohayani's exploration contribute valuable insights to the interplay between cryptography and steganography, presenting a pragmatic approach to fortify security while maintaining an inconspicuous facade in encrypted communication.

TEXT ENCRYPTION: HYBRID CRYPTOGRAPHIC METHOD USING VIGENERE AND HILL CIPHERS

A study conducted by Hamza, T. et al. (2020) highlights the historical role of cryptography in ensuring confidentiality for secret communications. However, with the substantial evolution of computer networks, cryptography has now become essential for preserving the privacy of digital information. Early encryption methods, such as the Caesar cipher and subsequent mono-alphabetic substitution encryption techniques, were groundbreaking but vulnerable to statistical attacks. In their 2020 research, TOUIL, H. et al. introduced a hybridization of Vigenere and Hill encryption techniques. This approach aimed to enhance the Vigenere algorithm by incorporating improvements from the Hill encryption method. Specifically, the focus was on addressing weaknesses related to detecting the key size for statistical attacks and the encryption susceptibility when two identical letters are encrypted in different blocks. The amalgamation of these methods resulted in a hybrid algorithm that demonstrates resilience against various attacks, including statistical assaults, presenting a dependable encryption solution.

Improved Approach of Rail Fence for Enhancing Security

A study conducted by Nahar & Chakraborty (2020) emphasizes the crucial role of encryption in transforming plain text into an unreadable format. While various encryption techniques exist, the Rail Fence technique stands out as one of the simplest and most captivating cryptographic algorithms. This method organizes plaintext in a crosswise fashion, allowing the production of ciphertext by reading from the top to the bottom row. However, this technique's limitation lies in its use of a restricted number of keys. Nahar, K., and Chakraborty, P. (2020), have innovatively enhanced the Rail Fence technique by introducing three fundamental phases: a substitution phase followed by two transposition phases. This novel approach effectively addresses the inherent weaknesses of the conventional Rail Fence technique by incorporating multiple encryption phases, thereby augmenting its strength and fortifying its resilience against potential vulnerabilities.

A Multilevel Approach of Transposition Ciphers for Data Security over Cloud

A research study by Thakkar & Thankachan (2021) addresses the ongoing concern of securing data stored in the cloud, a critical issue in contemporary times often tackled through diverse encryption techniques. To enhance confidentiality in communications, a variety of cryptographic encryption methods are employed. This study suggests an approach that utilizes a multilayer system with the goal of augmenting security measures. While the Rail Fence Cipher and the Simple Columnar Cipher each contribute to security independently, their individual use may provide limited protection, rendering the scheme somewhat decipherable. However, by sequentially combining these two ciphers, an additional layer of security is introduced. This multilayered method offers two tiers of encryption, significantly improving security measures and increasing the complexity for potential decryption attempts.

IMPROVING ONE-TIME PAD ALGORITHM ON SHAMIR'S THREE-PASS PROTOCOL SCHEME BY USING RSA AND ELGAMAL ALGORITHMS

A research conducted by Sidik et al. (2019) aims to address limitations identified in the one-time pad method of the Shamir Three-Pass Protocol. The primary focus is on correcting an inherent flaw within the one-time pad algorithm by modifying the ciphertexts generated through the three pathways of the protocol. This correction involves a two-step encryption process, where the ciphertext undergoes encryption using the RSA and ElGamal methods, resulting in a super ciphertext. The RSA encrypts the first and third lines, while the ElGamal method encrypts the second line, aligning with the protocol's objective of maintaining secrecy in key exchange.

The study's findings indicate the feasibility of rectifying the flaws in the one-time pad algorithm. However, it observes an increase in processing time as the length of the plaintext extends, nearing a ratio of one. Additionally, the successful decryption of super ciphertexts into the original three ciphertexts becomes crucial for deciphering the plaintext, presenting

a challenge, especially with the utilization of substantial prime numbers in the RSA and ElGamal algorithms.

Synthesis

Our thesis merges two classic encryption methods, the Caesar cipher and XOR operations, to form a more solid five-levels encryption and decryption algorithm. This innovation aims to significantly enhance data security and privacy in an increasingly vulnerable digital landscape. What sets our thesis apart is its logical approach. While many studies often focus solely on technical difficulties, our emphasis lies in creating a user-friendly application that demonstrates the real-world application of this refined algorithm. This is not just about introducing a new encryption algorithm; it is about offering a comprehensive solution for stronger data protection measures.

By blending established encryption methods, our thesis is not just contributing to existing systems, it is aiming to set new standards in data security. The fusion of the Caesar cipher and XOR operations into a five-level encryption process showcases a dedication to fortifying data security in a manner that is both innovative and practical. Moreover, the user-friendly design of our algorithm is a crucial aspect. Ensuring accessibility and ease of use across various fields and user profiles ensures that this advanced encryption technology is not limited to experts but can be embraced and utilized by a broader audience. This democratization of solid encryption is pivotal in today's data-driven world, where privacy and security are paramount concerns for individuals and organizations.

Our thesis doesn't just propose a concept, it's a tangible solution that bridges the gap between encryption techniques and practical implementation. It's not just about creating another encryption method, it's about empowering users with a powerful yet user-friendly tool to safeguard their sensitive information effectively.

Methodology

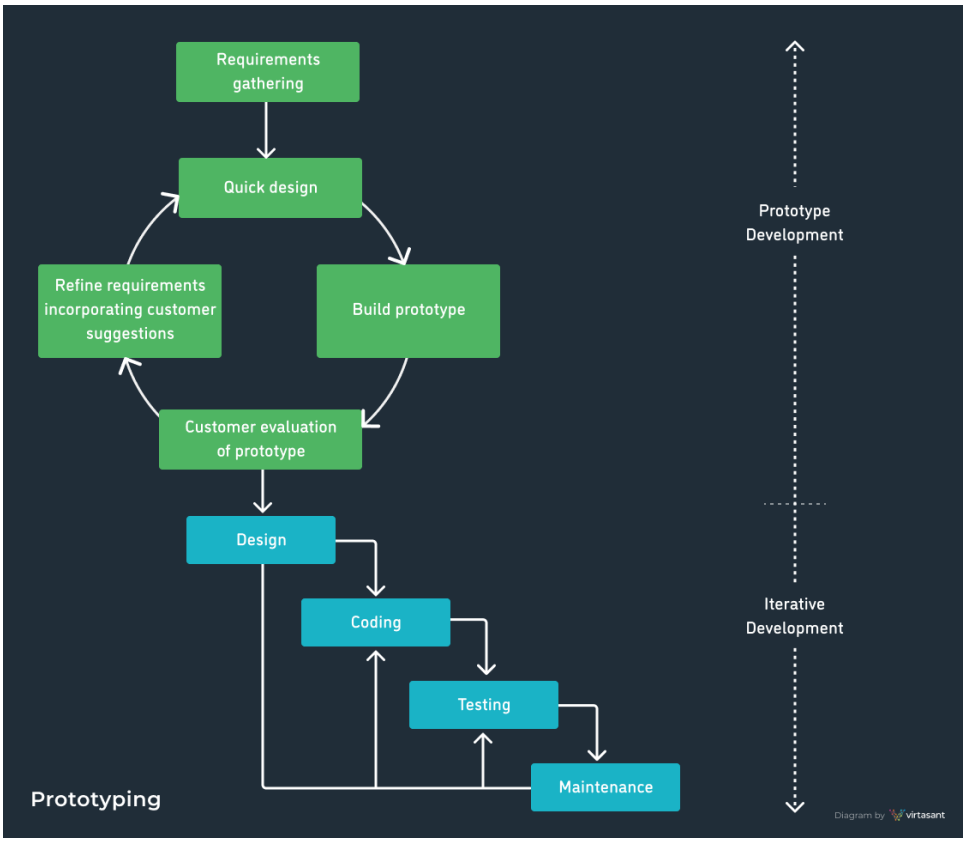


Figure 2: The Prototyping Model

Requirements gathering and analysis:

Requirement analysis is the first step in developing a prototyping model. During this phase, the system's desires are precisely defined. During the method, system users are interviewed to determine what they expect from the system.

Quick design:

The second phase could consist of a preliminary design or a quick design. During this stage, the system's basic design is formed. However, it is not a complete design. It provides the user with a quick overview of the system. The rapid design aids in the development of the prototype.

Build a Prototype:
During this stage, an actual prototype is intended to support the knowledge gained from quick design. It is a small low-level working model of the desired system.

Customer evaluation of the prototype:
The proposed system is presented to the client for preliminary testing at this stage. It is beneficial to investigate the performance model's strengths and weaknesses. Customer feedback and suggestions are gathered and forwarded to the developer.

Refine requirement of incorporating customer suggestion:
If the user is dissatisfied with the current model, you may want to improve the type that responds to user feedback and suggestions. When the user is satisfied with the upgraded model, a final system based on the approved final type is created.

Design:
In this phase, the software design is created, which includes the overall architecture of the software, data structures, and interfaces.

Coding:
The design is then implemented in code, usually in several iterations, and this phase is also called Development.

Testing:
The software is thoroughly tested to ensure that it meets the requirements and works correctly.

Maintenance:
This phase includes ongoing support, bug fixes, and updates to the software.

Overview of the project

Encryption Process

The encryption process begins with the program accepting an input text as plain text. The plain text is then subjected to a series of transformations, starting with passing it through the level 1 class encryption method. Following this, the plain text undergoes an addition of a prefix and suffix and is further extended. The extended text is then shifted based on the Caesar shift, resulting in a Caesared text. The Caesared text is converted into binary format. The converted binary text is subjected to XOR operations using the first XOR key, yielding the first encrypted text.

The first encrypted text becomes the input for the level 2 class encryption method, where it undergoes another XOR operation with the second XOR key, producing the second encrypted text. This process repeats through subsequent levels, with each encrypted text serving as input for the next level's encryption method and undergoing XOR operations with the corresponding XOR key. The sequential encryption continues until the fifth encrypted text is generated.

Upon obtaining the fifth encrypted text, an additional step is introduced to convert it to URL-safe Base64 encoding. This step is included for data transmission purposes, especially in web-based applications or scenarios where URLs are involved. The final output is the encoded fifth encrypted text, representing the result of the comprehensive encryption process.

Decryption Process

In the decryption process, the program begins by accepting an encrypted text. This encrypted text is then inputted into the level 5 class along with its corresponding decryption method. Following this, the encrypted text undergoes decoding with URL-safe Base64 decoding, converting it to its original binary format. The binary representation of the encrypted text is subjected to XOR operations using the fifth XOR key, resulting in the generation of the decrypted text five.

The decrypted text five becomes the input for the level 4 class, where another XOR operation occurs, utilizing the fourth XOR key, and returns as the decrypted text four. This decryption process repeats through subsequent levels, involving XOR operations with different keys. The decrypted text four is then inputted to the level 3 class, and the pattern repeats with XOR operations using the third XOR key to produce the decrypted text three. This sequential flow continues through levels 2 and 1 until the original text is reconstructed.

In the final steps of decryption, the decrypted text two is XORed with the first XOR key, and the result is converted back to ASCII characters. Following this, the converted ASCII characters undergo a Caesar shift, and the shifted text is sliced based on its prefix and suffix, ultimately resulting in the generation of the decrypted text one. The output of this process is the fully decrypted text, effectively reversing the encryption steps and revealing the original content.

System design specification

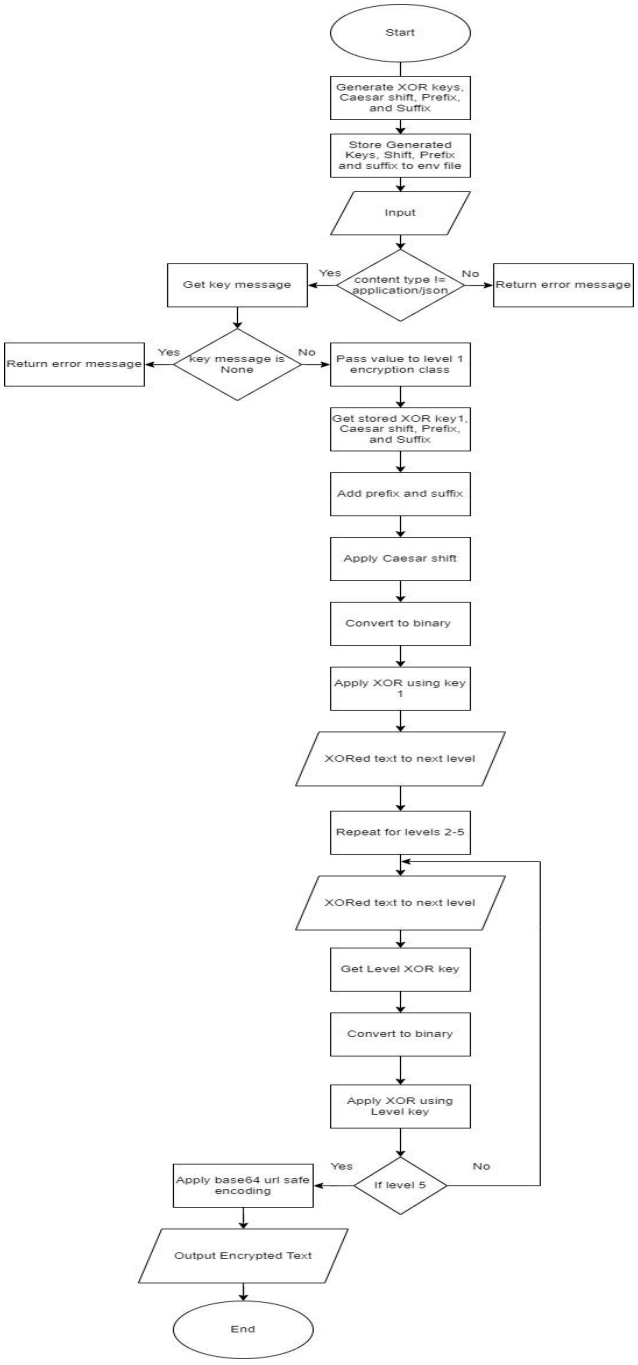


Figure 3: Encryption Flowchart

Commencing with the start shape, the Flask application begins its execution by autonomously generating XOR keys, Caesar shift values, prefixes, and suffixes. These

generated parameters are then stored in the environment (env) file. Upon receiving input, the application validates the content type, returning an error message if it is not application/json. Conversely, for application/Json content type, the key message is retrieved. Subsequently, the application checks if the key message is none, and if so, an error message is returned. If a key message is present, its value is then transmitted to the Level 1 encryption class.

In the Level 1 encryption process, the application accesses the XOR key, Caesar shift, prefix, and suffix stored in the env file for Level 1. The input text undergoes preprocessing, involving the addition of the prefix and suffix, followed by a Caesar shift operation. The shifted text is then converted to binary, and the resulting binary text undergoes XOR operation using the Level 1 XOR key.

The encryption process then proceeds to subsequent levels (Level 2 to Level 5) recursively. At each level, the input text is the XORED text from the preceding level. The application retrieves the stored XOR key for the respective level from the env file, converts the XOR key from hexadecimal to binary, and applies the XOR operation. This recursive encryption process continues until Level 5.

Before the final output, the XORED text from Level 5 undergoes additional processing: it is encoded using Base64 URL-safe encoding. The resulting encoded output represents the ultimate outcome of the encryption process. The flowchart concludes at this point, signifying the end of the entire process.

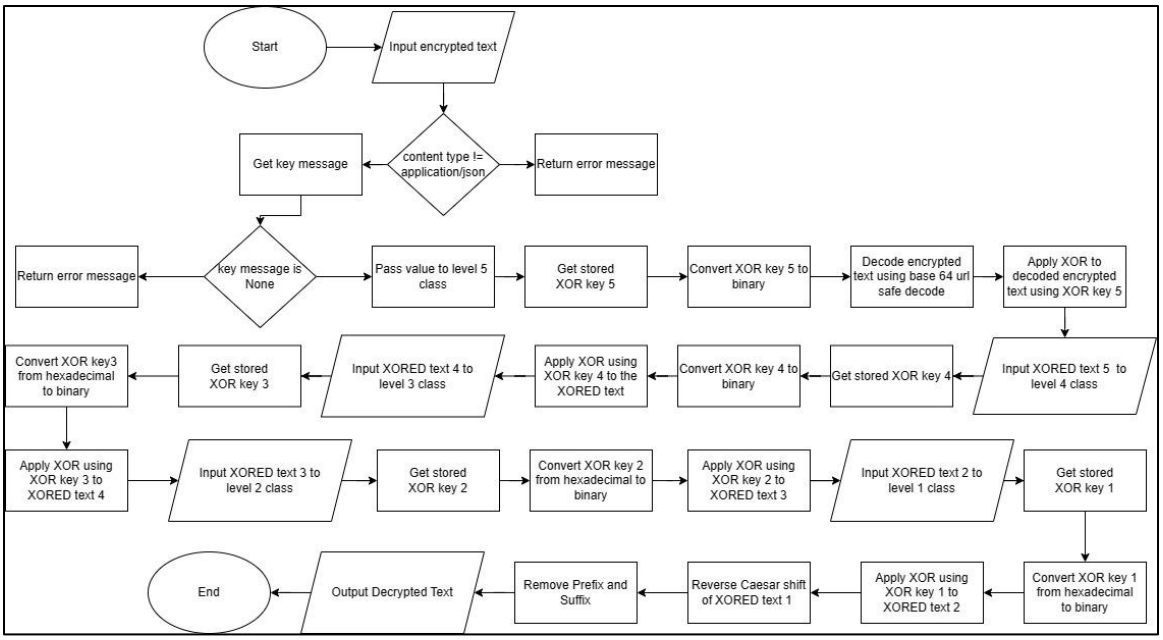


Figure # 4 Decryption Flowchart

The decryption process initiates with a start shape, where it takes encrypted text as input. Subsequently, the content type is verified to be application/json; if not, an error message is returned. However, if it is indeed application/json, the key message is obtained. It then checks if the key message is none; if so, an error message is returned. In the presence of a key message, the value is passed to the Level 5 decryption class.

The decryption flow continues as follows: the stored XOR key for Level 5 is retrieved, and it is converted to binary. The encrypted text undergoes Base64 URL-safe decoding, and the XOR key for Level 5 is applied to the decoded result. The resulting XORED text at Level 5 becomes the input for the Level 4 decryption class, where the process is repeated: the XOR key for Level 4 is retrieved, converted to binary, and used to perform XOR on the XORED text. This iterative process continues through Level 1.

Upon reaching Level 1, the XORED text is subjected to a Caesar shift in reverse to undo the original shift. Following that, the prefix and suffix are removed, and the decrypted text is outputted, marking the end of the decryption flowchart.



Figure 5: ERD Diagram

The Entity-Relationship Diagram (ERD) provided offers a profound insight into the intricate mechanics of a messaging application, unveiling a narrative of seamless communication, encryption, and system resilience. At its core, the diagram reflects the interconnectedness of user actions and system responses, mirroring the dynamics of human interaction in digital spaces. Initiating from the "Start" node, akin to the commencement of a conversation, users embark on their messaging journey. As messages unfold, the system diligently organizes them within the "Build Message List," akin to preparing thoughts before expression. However, the pivotal juncture emerges at "Encrypt Text?" echoing the human instinct to safeguard personal exchanges. Here, the system deliberates over the necessity of encryption, mirroring our inclination towards privacy and security. Through the lens of "Server Down," the diagram acknowledges the fragility of technology, reminiscent of unforeseen disruptions in human discourse. Yet, resilience shines through as the system navigates through challenges, ensuring message continuity and system robustness. Moreover, decryption processes unveil a narrative of understanding and connection, akin to deciphering the nuances of human expression. Through one-to-many and one-to-one relationships, the diagram captures the essence of interpersonal dynamics, illuminating the complexity and richness of digital communication landscapes. In essence, the ERD transcends its technical confines, offering a poignant reflection of human experiences and aspirations within the realm of modern technology.

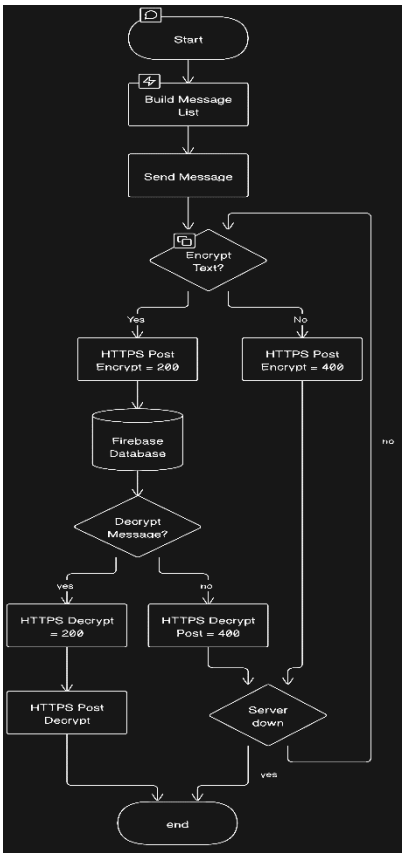


Figure 6: Dart Flowchart

In the realm of chat applications, the dynamic partnership between Flutter and Flask emerges as a beacon of modern app development. Flutter, the face of the frontend, crafts a user-centric experience with its intuitive widgets and fluid design. Through elements like ChatPage, users navigate seamlessly, composing and receiving messages with ease, thanks to Flutter's reactive architecture. On the backend, Flask serves as the engine, driving encryption, decryption, and data management with finesse. Its endpoints act as gateways, ensuring secure message transmission while storing data reliably in Firebase. This symbiotic relationship between frontend and backend epitomizes the essence of collaboration in technology, where user experience and data security converge harmoniously. Together, Flutter and Flask create an ecosystem where users connect and communicate effortlessly. It's a testament to the power of synergy, where frontend elegance meets backend resilience, enriching digital interactions with every exchange. In the ever-

evolving landscape of app development, this partnership stands as a testament to innovation, paving the way for immersive, secure, and seamless chat experiences.

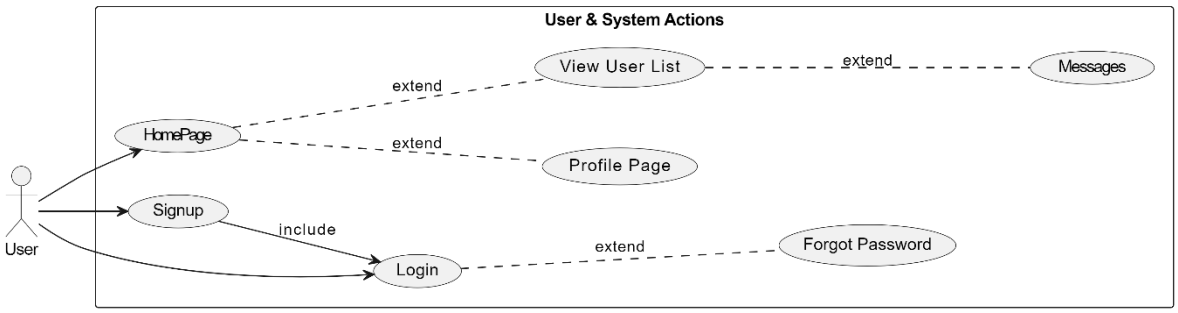


Figure 7: Use Case for Application/System

The Use Case Diagram reveals a multifaceted user experience within the messaging application. Users can seamlessly navigate through signup and login processes, leveraging secure authentication mechanisms to access the platform. Upon login, users gain access to various features, including profile management and encrypted messaging. The encryption process ensures that messages remain confidential during transmission, mitigating the risk of unauthorized access. Moreover, users can engage in collaborative communication by viewing decrypted messages, fostering a sense of transparency and trust within the messaging community. The Use Case Diagram illustrates the interactions between users and the system components in a messaging application. Users can navigate through various pages like signup, login, and profile pages, facilitated by the system. Upon logging in, users can access the chat page to send and receive encrypted messages. The system encrypts and decrypts messages for secure communication. Additionally, users can view a list of other users and initiate chats with them. The profile page allows users to view and edit their personal details such as username and bio. The system updates user information in the Firestore database. Another user can view decrypted messages, enhancing the collaborative aspect of the application. Overall, the diagram encapsulates the functionalities and user interactions within the messaging application, facilitating communication and user management effectively.

Hardware/Software

Hardware:

Mobile Phone – When our algorithm is implemented on the Developers application or system, the users must install the application to use the Developers app.

Computer – Same as the mobile phone to use the Developers system with our AC3 algorithm, they need to install the system on their computer.

Router – The application needs a router so the users can use the Developers application or the system.

Software:

Python – This is the programming languages that we use to create our AC3 Encryption and Decryption Algorithm and API

Dart - This is the programming we use to our frontend

Firebase – This is the programming that we use as hosting, and database to the chat

AWS – This is where the python running at EC2 Ubuntu Server 22.04.3 LT

RESULTS AND DISCUSSIONS

Results

This chapter will present the findings obtained from assessing the software and hardware of the AC3 Algorithm and the Demo Application. The evaluation involved testing and evaluating the capabilities and security of both the hardware and software systems, focusing on user experience. The intended respondents for rating and providing feedback on the researcher's system simulation were 13 developers. The gathered results will be discussed and analyzed alongside the collected data to substantiate each finding:

Table 1. Likert's Scale

Scale	Range	Interpretation
4	3.51 - 4.0	Excellent
3	2.51 - 3.50	Very Good
2	1.51 - 2.50	Fair
1	1.0 - 1.50	Poor

Evaluation of 13 IT Developer

Table 2. Functionality of the System

Functionality	mean	Interpretation
THE SYSTEM IS EASY TO OPERATE	3.7	Excellent
THE SYSTEM PROVISIONS FOR CONFORT AND CONVENIENCE	3.5	Very Good
THE SYSTEM OFFERS A SEAMLESS AND INTUITIVE USER EXPERIENCE	3.6	Excellent
THE SYSTEM ENHANCES EFFICIENCY AND PRODUCTIVITY IN TASKS OR PROCESSES	3.6	Excellent
THE SYSTEM PROVIDES TIMELY AND ACCURATE INFORMATION OR DATA	3.6	Excellent
General Average Mean	3.6	Excellent

The General Average Mean for functionality is calculated to be 3.6, which also falls under the "Excellent" interpretation. This suggests that, overall, respondents were highly satisfied with the system's functionality, finding it easy to operate, comfortable, and efficient, with a seamless user experience and reliable data provision.

Table 3. Reliability of the System

Reliability	Mean	Interpretation
THE SYSTEM PROVIDES RELIABLE INFORMATION	3.5	Very Good
THE SYSTEM PROVIDES ACCURATE REPORTS	3.7	Excellent
THE SYSTEM HAS ACCURACY IN PERFORMANCE	3.7	Excellent
THE SYSTEM HAS CONFORMANCE TO DESIRED OUTPUT	3.7	Excellent
General Average Mean	3.7	Excellent

Overall, the reliability of the system is rated highly, with mean scores ranging from 3.5 to 3.7, all falling within the "Very Good" to "Excellent" interpretations. This suggests that users perceive the system as reliable, accurate, and consistent in its performance and output, enhancing trust and confidence in its reliability.

Table 4. Efficiency of the System

Efficiency	Mean	Interpretation
THE SYSTEM PERFORMED ACCORDING TO THE SPECIFICATION	3.6	Excellent
THE SYSTEM RUNS EFFICIENTLY AND ON TIME	3.6	Excellent
THE SYSTEM RUNS AS EXPECTED	3.6	Excellent
THE SYSTEM HAS SECURITY PROVISION REQUIREMENTS	3.7	Excellent
General Average Mean	3.7	Excellent

The General Average Mean for efficiency is calculated to be 3.7, falling under the "Excellent" interpretation. Overall, users rated the system's efficiency highly, with mean scores ranging from 3.6 to 3.7, indicating that the system performed according to specifications, ran efficiently and securely, meeting user expectations consistently.

Table 5. Portability of the System

Portability	Mean	Interpretation
THE SYSTEM IS EASY TO INSTALL	3.9	Excellent
THE SYSTEM ADAPTS TO SOFTWARE REQUIREMENTS	3.5	Very Good
THE SYSTEM IS MANAGEABLE TO OPERATE	3.6	Excellent
THE SYSTEM IS EASY TO USE, ESPECIALLY FOR THE BEGINNER OF THE APPLICATION	3.7	Excellent
General Average Mean	3.7	Excellent

The General Average Mean for portability is calculated to be 3.7, which falls under the "Excellent" interpretation. Overall, users rated the system's portability highly, with mean scores ranging from 3.5 to 3.9, indicating that the system is easy to install, manage, and use, with good adaptability to software requirements.

Table 6: Overall

Category	Mean	Interpretation
Functionality	3.6	Excellent
Reliability	3.7	Excellent
Effeciency	3.7	Excellent
Portability	3.7	Excellent
General Average Mean	3.7	Excellent

The General Average Mean across all categories is calculated to be 3.7, falling under the "Excellent" interpretation. Overall, users rated the system highly across all aspects, indicating high levels of satisfaction with its functionality, reliability, efficiency, and portability.

Findings

The evaluation of the AC3 Algorithm and the Demo Application involved a total of 13 Developers, whose insights provided valuable feedback on the system's performance. Across the board, the system received commendable ratings, with an average score of 3.7 in functionality, reliability, efficiency, and portability. Notably, users appreciated the system's functionality, finding it intuitive and efficient, while also praising its reliability in delivering accurate information and reports. Furthermore, the system's efficiency and portability were lauded for their seamless performance and ease of adaptation to varying software requirements. This comprehensive assessment, involving both IT specialists and Developers, underscores the system's robustness and suitability for diverse user needs, cementing its position as a reliable solution in the technological landscape.

CONCLUSIONS AND RECOMMENDATIONS

Conclusion

Considering the objectives outlined in this study, which focused on the development, implementation, and evaluation of encryption and decryption solutions integrating the Caesar cipher and XOR operations, it can be conclusively stated that the study has successfully achieved its aims. Through the meticulous design and execution of an innovative encryption and decryption algorithm, the study sought to bolster data security and privacy, providing a potent tool for safeguarding sensitive information.

The assessment of the AC3 Algorithm and the Demo Application yielded commendable results, indicating a resounding success in meeting the study's objectives. With an average rating of 3.7 across functionality, reliability, efficiency, and portability, the system garnered praise for its intuitive operation, reliability in delivering accurate information, seamless performance, and adaptability to varying software requirements. The high level of satisfaction expressed by both IT specialists and Developers underscores the robustness and efficacy of the system, affirming its relevance in addressing contemporary data security challenges.

Recommendation

Building upon the achievements of this study, several recommendations are proposed to further enhance the project's efficiency and effectiveness.

- **Continuous Security Enhancements:** Despite the commendable ratings for security, it's imperative to stay proactive in addressing emerging threats. Implement regular updates and security patches to fortify the system against evolving cybersecurity risks.
- **Enhanced User Experience:** Invest in refining the user interface of the Demo Application to ensure an even more intuitive and seamless experience for developers and end-users alike. Incorporate user feedback to streamline workflows and enhance usability.
- **Scalability and Compatibility:** Explore opportunities to enhance the system's scalability and compatibility with a wider range of platforms and software environments. This would extend its utility and accessibility to a broader audience of developers and organizations.

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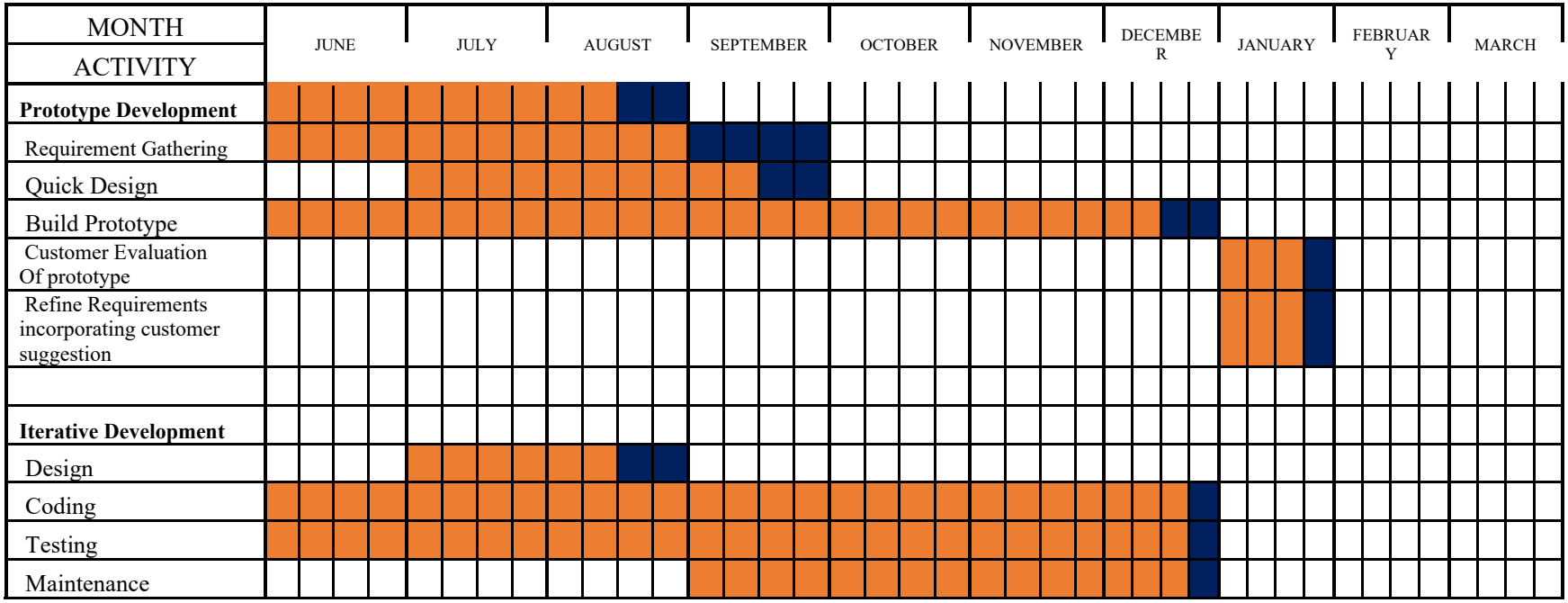
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APPENDICES

APPENDIX A. GANTT CHART



Legend:

Orange is Ongoing

Blue is Done

APPENDIX B. ACTUAL THESIS EXPENSES

THESIS EXPENSES

Quantity	Specifics	Actual Cost
1	AWS Yearly cost	10php
1	Domain year	120php
1	SSL certificate	430php
1	Firebase database & Hosting	Free

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APPENDIX C. USER'S MANUAL

Appendix C. User's Manual

Insert here the pdf copy of your User's Manual. You may use one or more pages for your user's manual. Ensure that all necessary information are included

Application:

1. When you finish downloading the app or redirect from the browser you will be greeted to the login page then navigate to signup button to create an account

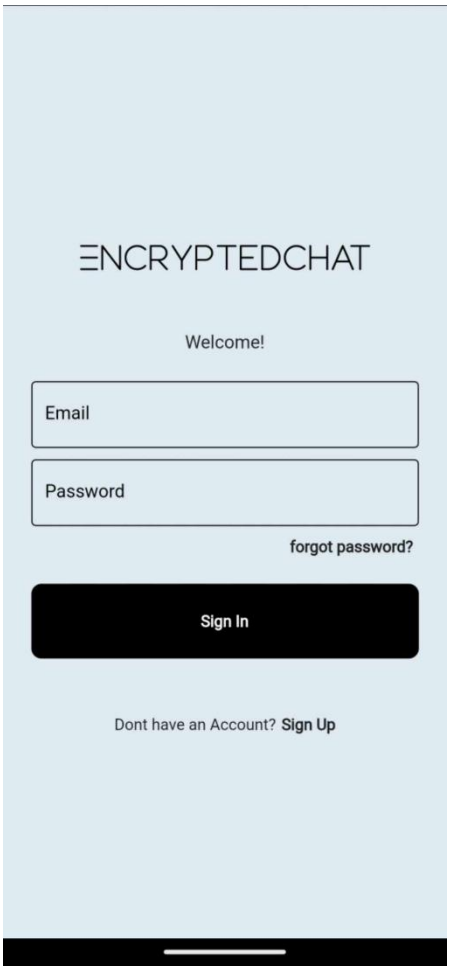


Figure 8: Login page

After created an account that you provide your information in the signup page login your account it you will be greeted to the HomePage

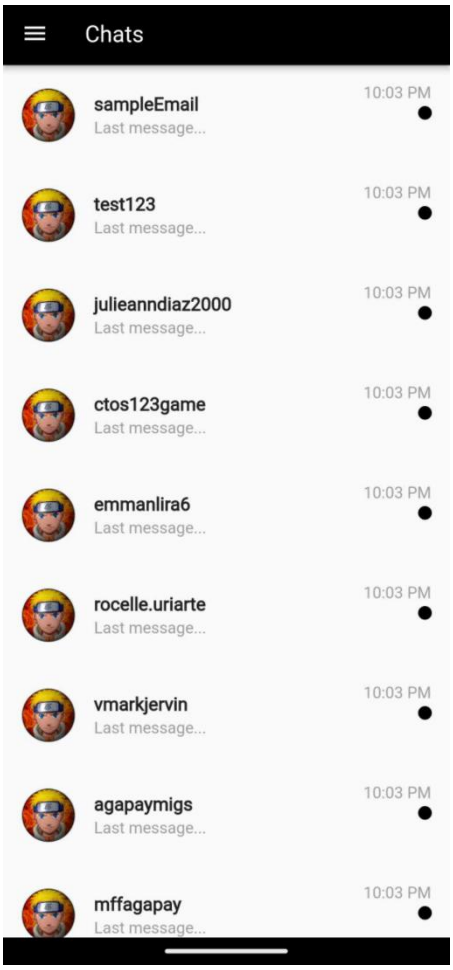


Figure 9: Home page

To check the user chatbox the specific user that you want to chat the after that you will be greeted of the messages of the user

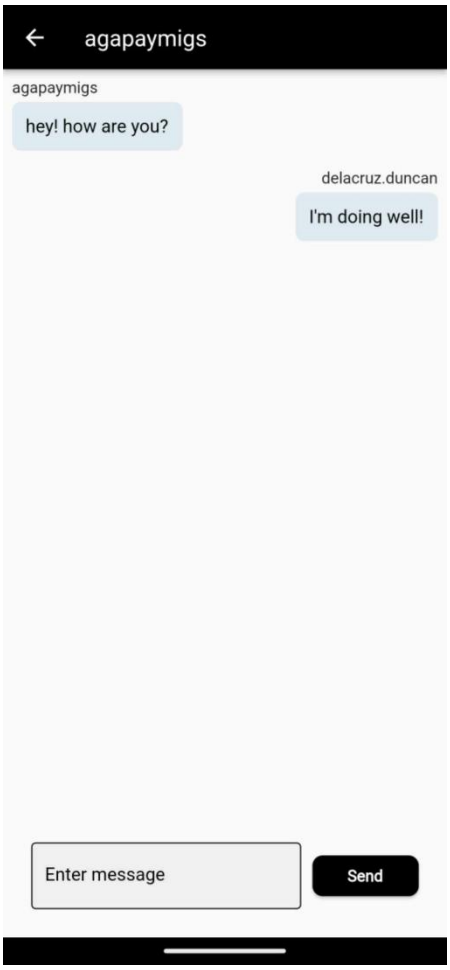


Figure 10: Messages

In the messages you can type the message that you want and just hit send when you are ready to send the message

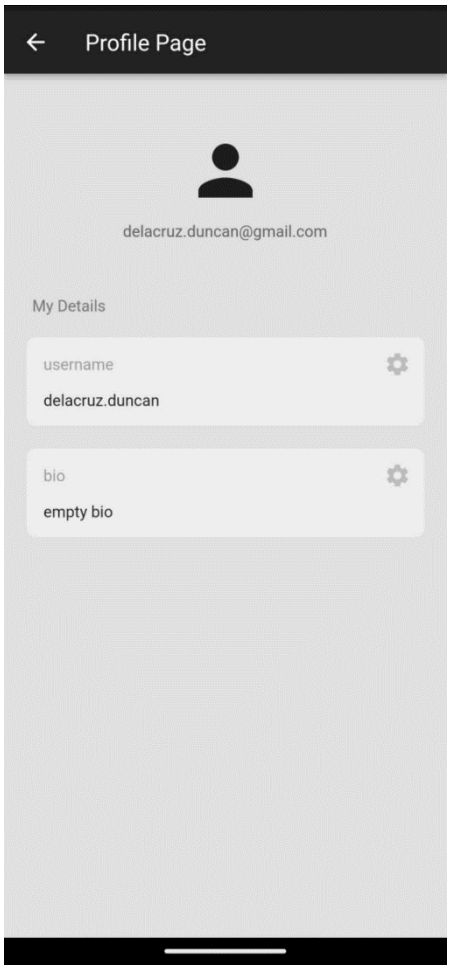
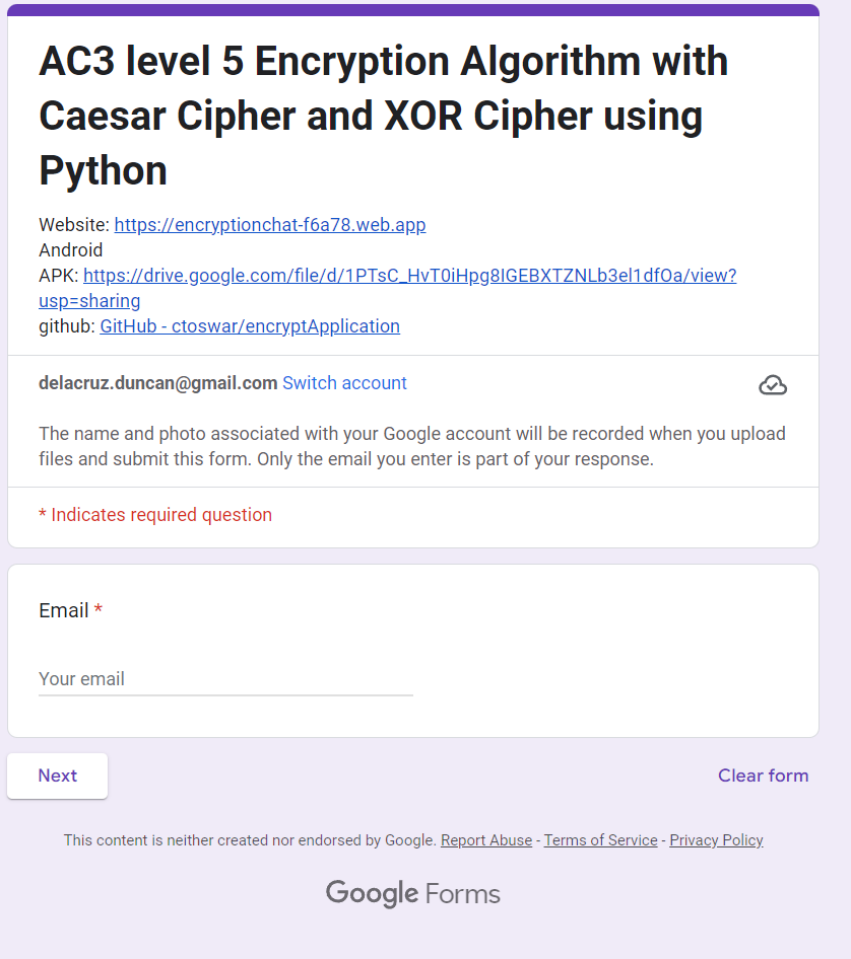
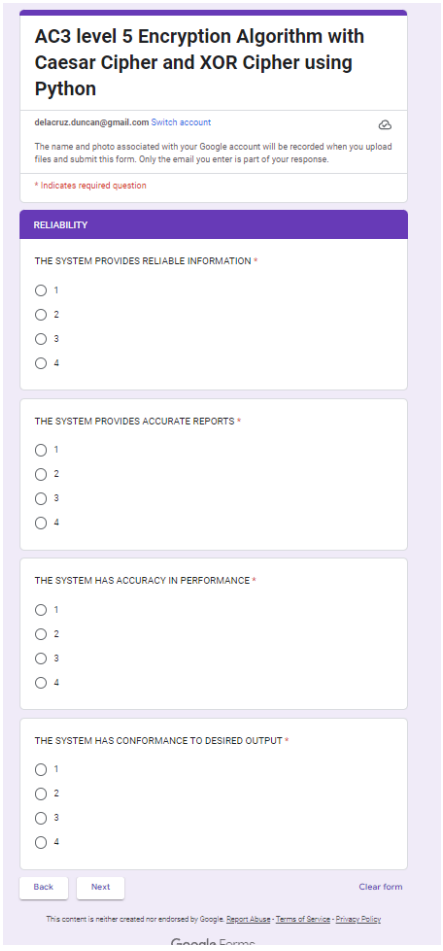


Figure 11: Profile Page

When you are in the HomePage still and you want to change your username Navigate to the Hamburger button (three line on the top left) then click profile you will be greeted the profile page then you can change your username and bio in there

APPENDIX D. SAMPLE EVALUATION


AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python
Website: https://encryptionchat-f6a78.web.app
Android
APK: https://drive.google.com/file/d/1PTsC_HvT0iHpg8IGEBXTZNLb3el1dfOa/view?usp=sharing
github: GitHub - ctoswar/encryptApplication
delacruz.duncan@gmail.com Switch account
The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.
* Indicates required question
Email *
Your email
Next
Clear form
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Google Forms



AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

delacruz.duncan@gmail.com Switch account

The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.

* Indicates required question

RELIABILITY

THE SYSTEM PROVIDES RELIABLE INFORMATION *

1
2
3
4

THE SYSTEM PROVIDES ACCURATE REPORTS *

1
2
3
4

THE SYSTEM HAS ACCURACY IN PERFORMANCE *

1
2
3
4

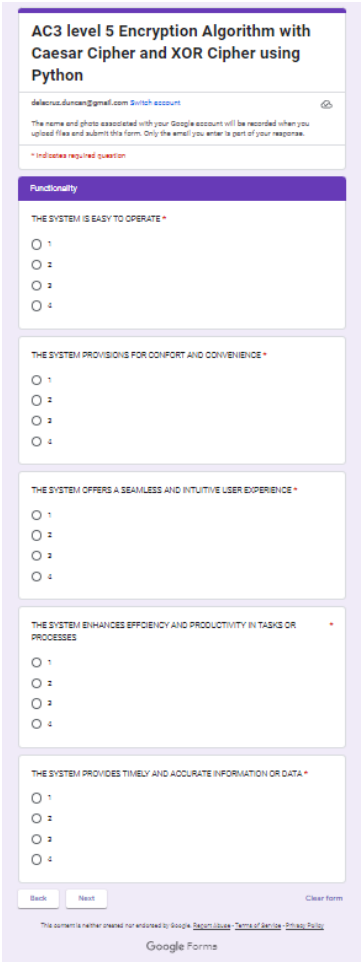
THE SYSTEM HAS CONFORMANCE TO DESIRED OUTPUT *

1
2
3
4

Back Next Clear form

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AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

delacruz.duncan@gmail.com Switch account

The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.

* Indicates required question

Functionality

THE SYSTEM IS EASY TO OPERATE *

1
2
3
4

THE SYSTEM PROVISIONS FOR COMFORT AND CONVENIENCE *

1
2
3
4

THE SYSTEM OFFERS A SEAMLESS AND INTUITIVE USER EXPERIENCE *

1
2
3
4

THE SYSTEM ENHANCES EFFICIENCY AND PRODUCTIVITY IN TASKS OR PROCESSES *

1
2
3
4

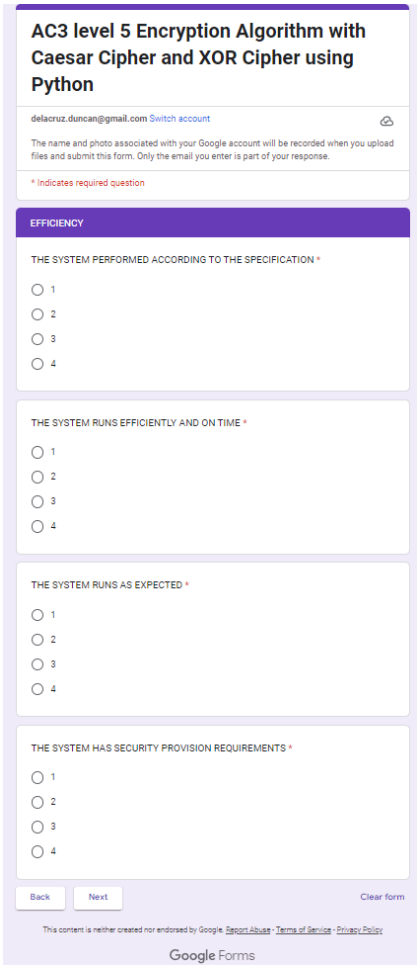
THE SYSTEM PROVIDES TIMELY AND ACCURATE INFORMATION OR DATA *

1
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AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

delacruz.duncan@gmail.com Switch account

The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.

* Indicates required question

EFFICIENCY

THE SYSTEM PERFORMED ACCORDING TO THE SPECIFICATION *

1
2
3
4

THE SYSTEM RUNS EFFICIENTLY AND ON TIME *

1
2
3
4

THE SYSTEM RUNS AS EXPECTED *

1
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3
4

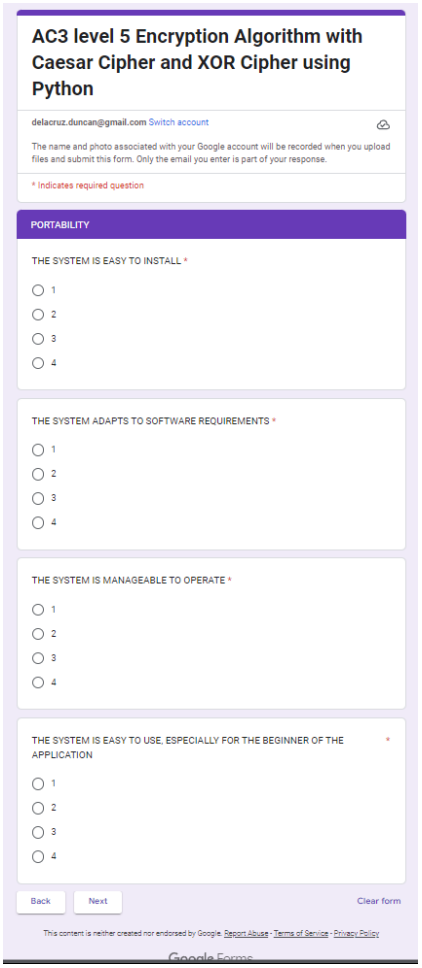
THE SYSTEM HAS SECURITY PROVISION REQUIREMENTS *

1
2
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4

Back Next Clear form

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AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

delacruz.duncan@gmail.com Switch account

The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.

* Indicates required question

PORTABILITY

THE SYSTEM IS EASY TO INSTALL *

1
2
3
4

THE SYSTEM ADAPTS TO SOFTWARE REQUIREMENTS *

1
2
3
4

THE SYSTEM IS MANAGEABLE TO OPERATE *

1
2
3
4

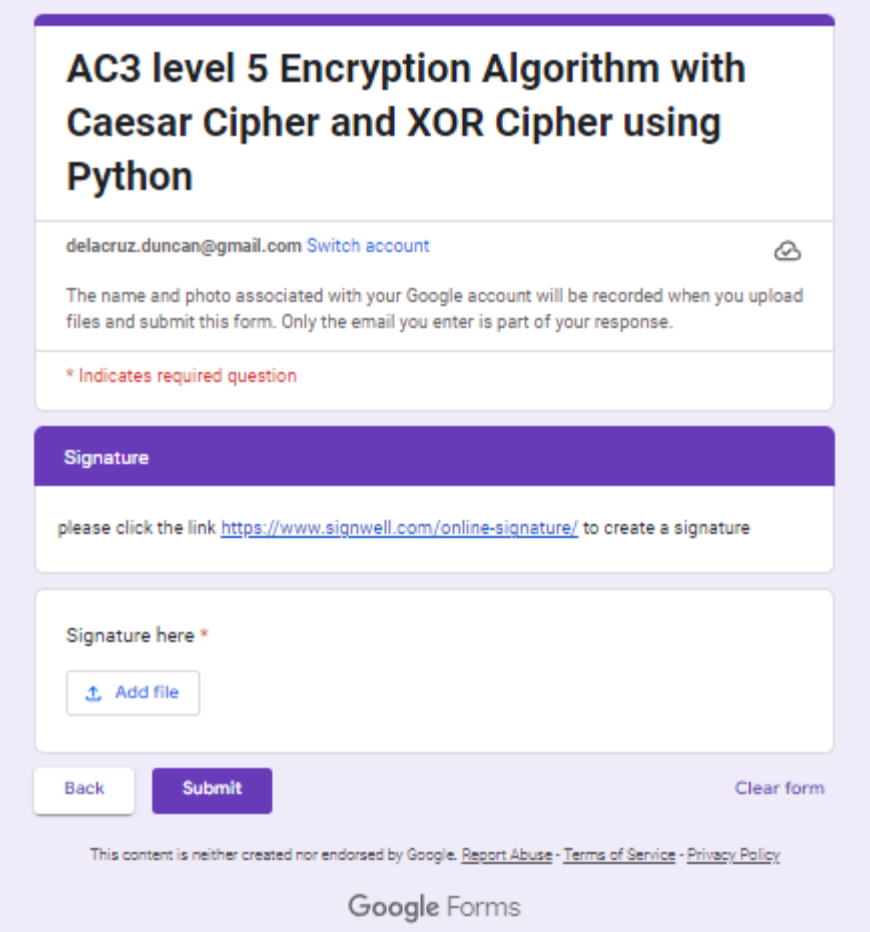
THE SYSTEM IS EASY TO USE, ESPECIALLY FOR THE BEGINNER OF THE APPLICATION *

1
2
3
4

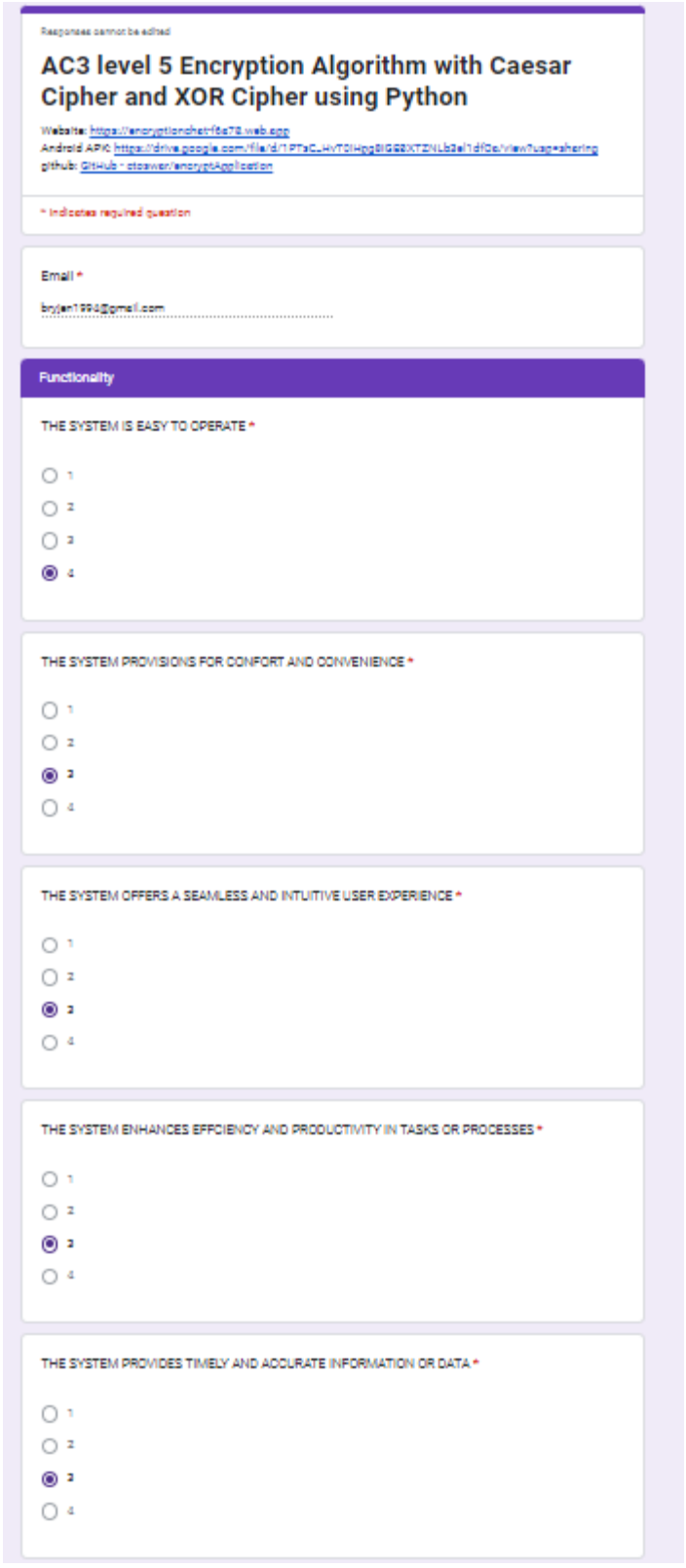
Back Next Clear form

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AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python
delacruz.duncan@gmail.com Switch account
The name and photo associated with your Google account will be recorded when you upload files and submit this form. Only the email you enter is part of your response.
* Indicates required question
Signature
please click the link https://www.signwell.com/online-signature/ to create a signature
Signature here *
Add file
Back
Submit
Clear form
This content is neither created nor endorsed by Google. Report Abuse - Terms of Service - Privacy Policy
Google Forms

APPENDIX E. COPY OF THE EVALUATION SHEETS OF THE RESPONDENT



Responses cannot be edited

AC3 level 5 Encryption Algorithm with Caesar Cipher and XOR Cipher using Python

Website: https://encryptionchat-f6e78.web.app
Android APK: https://drive.google.com/file/d/1PTaCJHVT0IHpg9IG99XTZiVLb3eI1dfCe/view?usp=sharing
github: GitHub - stoawer/encryptApplication

* Indicates required question

Email *

bryan1994@gmail.com

Functionality

THE SYSTEM IS EASY TO OPERATE *

- 1
- 2
- 3
- 4

THE SYSTEM PROVISIONS FOR CONFORT AND CONVENIENCE *

- 1
- 2
- 3
- 4

THE SYSTEM OFFERS A SEAMLESS AND INTUITIVE USER EXPERIENCE *

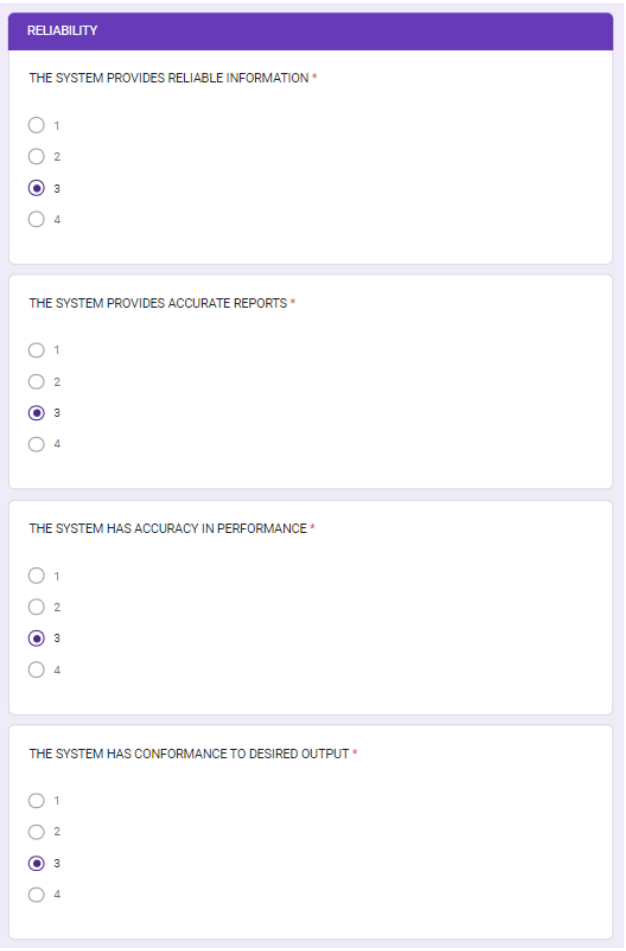
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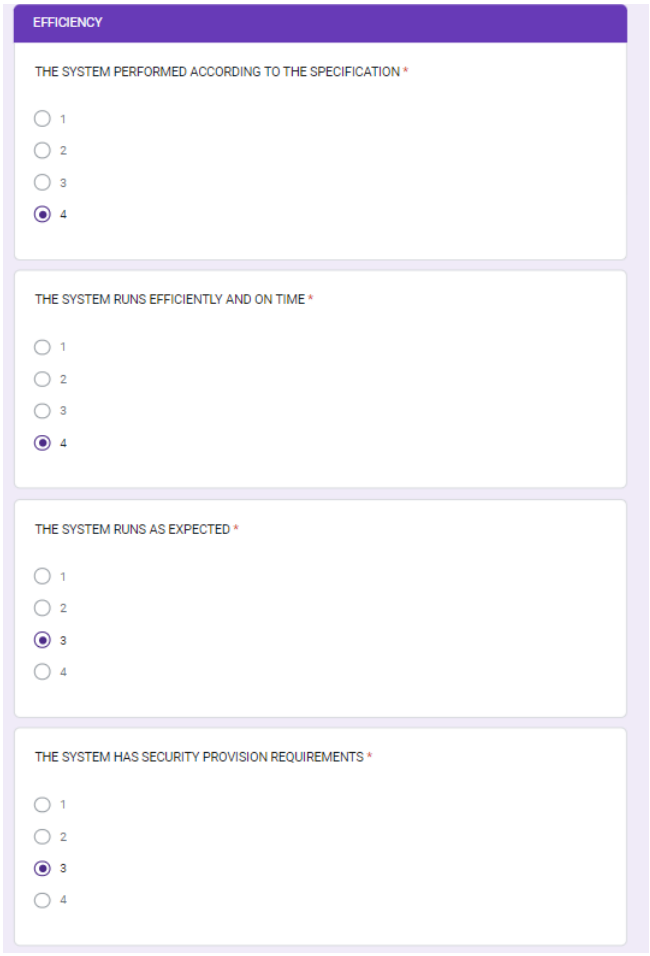
THE SYSTEM ENHANCES EFFCIENCY AND PRODUCTIVITY IN TASKS OR PROCESSES *

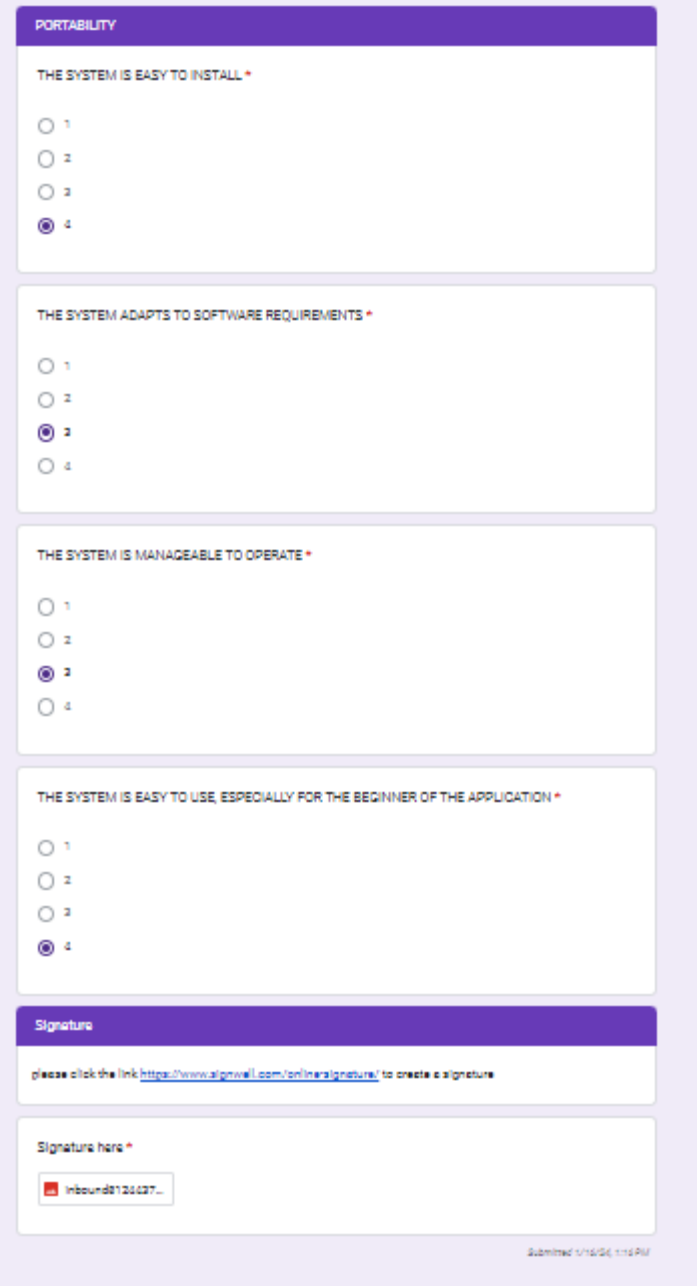
- 1
- 2
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- 4

THE SYSTEM PROVIDES TIMELY AND ACCURATE INFORMATION OR DATA *

- 1
- 2
- 3
- 4


RELIABILITY
THE SYSTEM PROVIDES RELIABLE INFORMATION *
1
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4
THE SYSTEM PROVIDES ACCURATE REPORTS *
1
2
3
4
THE SYSTEM HAS ACCURACY IN PERFORMANCE *
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2
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4
THE SYSTEM HAS CONFORMANCE TO DESIRED OUTPUT *
1
2
3
4


EFFICIENCY
THE SYSTEM PERFORMED ACCORDING TO THE SPECIFICATION *
1
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4
THE SYSTEM RUNS EFFICIENTLY AND ON TIME *
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2
3
4
THE SYSTEM RUNS AS EXPECTED *
1
2
3
4
THE SYSTEM HAS SECURITY PROVISION REQUIREMENTS *
1
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PORTABILITY
THE SYSTEM IS EASY TO INSTALL *
1
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4
THE SYSTEM ADAPTS TO SOFTWARE REQUIREMENTS *
1
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THE SYSTEM IS MANAGEABLE TO OPERATE *
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THE SYSTEM IS EASY TO USE, ESPECIALLY FOR THE BEGINNER OF THE APPLICATION *
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Signature
please click the link https://www.signwell.com/online-signature/ to create a signature
Signature here *

APPENDIX F. SOURCE CODE

Python Source Code

```
@staticmethod
def _apply_caesar_shift(text, shift):
    caesar_text = "".join(chr((ord(char) - 32 + shift) % 96 + 32) for char in text)
    return caesar_text

@staticmethod
def _apply_xor(text, xor_key):
    text_bytes = text.encode('utf-8')
    encrypted_bytes = [text_bytes[i] ^ xor_key[i % len(xor_key)] for i in
range(len(text_bytes))]
    return bytearray(encrypted_bytes)

def encrypt(self, input_text):
    plain_text = self.__prefix + input_text + self.__suffix
    caesar_text = self._apply_caesar_shift(plain_text, self.__caesar_shift)
    encrypted_text = self._apply_xor(caesar_text, self.__xor_key)
    return encrypted_text

def decrypt(self, encrypted_text):
    decrypted_bytes = bytearray()
    for i, byte in enumerate(encrypted_text):
        decrypted_byte = byte ^ self.__xor_key[i % len(self.__xor_key)]
        decrypted_bytes.append(decrypted_byte)

    caesar_text = "".join(chr(byte) for byte in decrypted_bytes)
    original_text = self._apply_caesar_shift(caesar_text, -self.__caesar_shift)
    decrypted_text = original_text[len(self.__prefix):-len(self.__suffix)]
    return decrypted_text

@staticmethod
def encrypt(plaintext_bytes, xor_key):

    encrypted_bytes = bytearray()
    for i, byte in enumerate(plaintext_bytes):
        encrypted_byte = byte ^ xor_key[i % len(xor_key)]
        encrypted_bytes.append(encrypted_byte)

    return encrypted_bytes

@staticmethod
def decrypt(encrypted_text, xor_key):

    decrypted_bytes = bytearray()
    for i, byte in enumerate(encrypted_text):
        decrypted_byte = byte ^ xor_key[i % len(xor_key)]
```

```
            decrypted_bytes.append(decrypted_byte)

        return decrypted_bytes


def __init__(self, level=2):
    xor_key_hex = os.getenv(f'XOR_KEY_LEVEL{level}')

    if xor_key_hex is None:
        raise ValueError(f'XOR key for level {level} not found in the environment.')

    self.__XOR_Key = bytes.fromhex(xor_key_hex)

def encrypt(self, encrypted_bytes):
    return XORCipher.encrypt(encrypted_bytes, self.__XOR_Key)

def decrypt(self, input_text):
    return XORCipher.decrypt(input_text, self.__XOR_Key)
```

```
def encrypt_message():
    try:
        data = request.get_json()
        message = data.get('message')

        logger.debug("Received request with message: %s", message)

        if not message:
            return jsonify({'error': 'Missing message parameter'}), 400

        encrypted_message = main_controller.encrypt(message)

        # Push encrypted message to Firebase
        push_to_firebase({'type': 'encrypt', 'message': message, 'encrypted_message':
encrypted_message})

        response = {
            'encrypted_message': encrypted_message
        }
        return jsonify(response), 200
    except Exception as e:
        logger.error("An error occurred: %s", str(e))
        return jsonify({'error': 'An error occurred'}), 500

def decrypt_message():
    try:
        data = request.get_json()
        encrypted_text = data.get('encrypted_text')

        logger.debug("Received request with encrypted text: %s", encrypted_text)

        if not encrypted_text:
            return jsonify({'error': 'Missing encrypted_text parameter'}), 400

        original_text = main_controller.decrypt(encrypted_text)

        # Push decrypted message to Firebase
        push_to_firebase({'type': 'decrypt', 'encrypted_text': encrypted_text,
'original_text': original_text})

        response = {
            'original_text': original_text
        }
        return jsonify(response), 200
    except Exception as e:
```

```
        logger.error("An error occurred: %s", str(e))
        return jsonify({'error': 'An error occurred'}), 500
```

Dart Source Code

```
if (_messageController.text.isNotEmpty) {
      // Encrypt the message before sending it
      final encryptedMessage = await _encryptMessage(_messageController.text);

      await _chatService.sendMessage(widget.receiverUserID, encryptedMessage);
      _messageController.clear();
    }
  }

  Future<String> _encryptMessage(String message) async {
    final response = await http.post(
      Uri.parse('$serverURL/encrypt'),
      headers: <String, String>{
        'Content-Type': 'application/json',
      },
      body: jsonEncode({'message': message}),
    );

    if (response.statusCode == 200) {
      return jsonDecode(response.body)['encrypted_message'];
    } else {
      throw Exception('Failed to encrypt message');
    }
  }

  Future<String> _decryptMessage(String encryptedMessage) async {
    final response = await http.post(
      Uri.parse('$serverURL/decrypt'),
      headers: <String, String>{
        'Content-Type': 'application/json',
      },
      body: jsonEncode({'encrypted_text': encryptedMessage}),
    );
```

APPENDIX G. CURRICULUM VITAE OF RESEARCHERS

Curriculum Vitae of

Miguel Florencio F. Agapay

Veraville, Dulong Bayan, Bacoor Cavite
mffagapay@gmail.com
09175281980



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2021-2024	STI College Bacoor
TechVoc	2018-2020	Far Eastern University Manila
High School	2017-2018	Ruther E. Esconde School of Multiple and Intellligences, Inc
Elementary	2009-2016	Ruther E. Esconde School of Multiple and Intellligences, Inc.

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2020	Work Immersion Program	Web Design Developer

Listed in reverse chronological order (most recent first).

SKILLS

UI/IX Designer
Frontend Developer
Java Programming

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
February 2020	freeCodeCamp Web Design Developer

Curriculum Vitae of

Duncan John D. Carpiso

Alder Street Lancaster Residence Imus Cavite
Delacruz.duncan@gmail.com
09273300854



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2020 - 2024	STI College Bacoor
TechVoc	2018-2020	Manuel A. Roxas HighSchool
High School	2014-2017	Manuel A. Roxas HighSchool
Elementary	2008-2014	Dr. Celedonio A. Salvador Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2019	Mariana Academy of Maritime	Frontdesk Computer Asst.

Listed in reverse chronological order (most recent first).

SKILLS

Java
Python
Dart
PHP
AWS
MySQL
GIT

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
Jan 2020	National Certificate II Computer Systems Servicing

Curriculum Vitae of

Carlo R. Cobey

Ugalde Comp, Panapaan 2, Bacoor City, Cavite
Carlocobey0204@gmail.com
09919166546



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2019 - 2024	STI College Bacoor
TechVoc	2017-2019	Montessori Professional College Imus
High School	2013-2017	Bacoor National Highschool Tabing-dagat
Elementary	2006-2013	Gov. P.f Espiritu Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2018 - Present	Crew Trainer	Mcdonald's new molino

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

Listed in reverse chronological order (most recent first).

SKILLS

Java

Python

C#

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop

Curriculum Vitae of

Leonavelle Marjorie Joi Camatog

1052 Patindig Araw Bayan Luma 8 , Imus Cavite
leona_cutee@yahoo.com
09282222763



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2021-2024	STI College Bacoor
	2018-2020	Sti College Bacoor
High School	2017-2018	Als Impals
Elementary	2004-2009	Imus Unida Christian School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

Listed in reverse chronological order (most recent first).

SKILLS

Java Programming

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates Title of Training, Seminar or Workshop